

Final Year Project at School of Mathematics & Physics
A Frame Work for Diffusion Process Approximation by
Continuous-Time Markov Chain
连续时间马氏链逼近扩散过程的标准框架

A THESIS

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Abstract

This report explores a general framework for approximating diffusion processes using the continuous-time Markov chain approximation (MCA) methodology. Building on previous MCA methods, we attempt to propose methodological criteria concerning operator convergence analysis and prove that the choice of discrete methods for state spaces in MCA corresponds to different levels of convergence accuracy. The report consists of three main parts: operator convergence conditions, error analysis with grid design, and innovations in parameter estimation algorithms for this framework.

本报告探究了一种使用连续时间马尔可夫链近似（MCA）方法来逼近一般扩散过程的标准框架。在以往MCA方法的基础上，我们尝试提出关于算子收敛分析的选择标准，并证明了在MCA方法中针对状态空间选择不同离散格点方法，会对应不同层次的收敛精度。此报告主要包括三个部分：算子收敛条件、格点设计的误差分析，以及在该框架下参数估计算法的创新。

KEY WORDS: Markov chain approximation; Diffusion process; Non-uniform grids; Convergence analysis; Parameter estimation

关键词：马尔可夫链近似；扩散过程；非均匀格点；收敛分析；参数估计

Acknowledgement

Thinking time was vast, yet unaware how swiftly it would slip away.

Two years ago, I first encountered the question in this article,
At that moment, confronted by the enigmatic term "Markov Chain",
I never imagined it would come to fill the entirety of my youth's domain.

Now, as I gaze at the spring outside the math building, my mind drifts,
Unlike a Poisson process that has awaited its next shift,
I long for a little more time, a moment to pause,
To write a few more lines in the hurried pages of youth's cause.

Gratitude to my teachers, friends, and family who have supported me;
you are the most beautiful memories of my whole life.
To mathematics,
you are the most splendid wave in my youth's sea.

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Chapter 1

Introduction

The motivation for the present work arises from the paper [9], where Cui, Kirkby, and Nguyen (2017) established a novel and imaginative continuous-time Markov chain approximation (CTMC) combined with a transform-based regime-switching model for derivative pricing. In their work, they were highly innovative in simulating the stochastic volatility process within the CTMC approximation framework and in using the state space values as estimates for the volatility parameters in the regime-switching model. During numerical implementation, the Markov chain requires a sufficiently fine discretization of the state space to accurately approximate a diffusion process. However, too many states can complicate the regime-switching model and make it difficult to handle. Thus, we face a trade-off between the number of states and the associated error. This article is inspired by this issue and proposes methodological criteria for grid design on the state space to achieve higher accuracy.

Diffusion processes evolve a continuous space in both time and state, which limits their solution in numerical computations for real-life cases. Therefore, reasonable numerical approximation methods are needed to obtain solutions to such equations. Two possible directions for achieving efficient and accurate approximations are:

- *Time Discretization:* Divide the time scale into a finite number of discrete grids, while keeping the state space continuously in the diffusion process. Then, use methods like the Euler scheme or the Milstein scheme to approximate how the diffusion process evolves over time.
- *State Discretization:* Divide the state scale into a finite number of discrete grids, while keeping the time dimension continuously in the diffusion process. Then, consider a diffusion approximation evolves a continuous-time Markov chain.

The time discretization method has been well-developed over the past decades and is the most widely used technique for solving stochastic differential equations (SDEs), such as the Black-Scholes equation, in various financial models. However, the CTMC approximation method is also a natural, novel and efficient tool to solve such problems, with many excellent numerical properties that worth further study.

The idea of using a continuous-time Markov chain to approximate diffusion processes was first developed by Kushner (1977) in [1] as an approach to solve optimal control and non-linear

filtering problems in diffusion models. Later, Piccioni (1987) in [2] proposed the CTMC approximation method, which uses the intensity rate matrix (or generator matrix) of a continuous-time Markov chain to characterize its transition probability. This is also, in fact, a discrete-state analogue to the infinitesimal generator of the diffusion process. However, this method is relatively crude as it only match the first moment of diffusion process. In [8], Lo and Skindilias (2014) improved the CTMC approximation method by developing a non-uniform discretization method for the state space and suggested a heuristic approach for determining the grid elements, which fitted the higher order moments of diffusion. Still, there left many different non-equidistant discretization methods for the state space unexplored. For instance, Tavella and Randall (2000) in [4] presented various discretization methods for differential equations that were used in [9]. How to choose the discretization method is still a problem to be studied. It is therefore worthwhile to further investigate the approximation principles and find a general criteria for CTMC approximation methods.

Mijatovic and Pistorius (2009) in [7] applied a similar CTMC approximation method to the pricing of Asian and exotic options. Their goal was to develop methods with fast convergence rates for specific financial models, which has the similar principles outlined by Lo and Skindilias (2014) in [8]. Furthermore, they proposed conclusions regarding the selection of the generator matrix and discussed special forms of the jump-diffusion process. Recently, several studies have further advanced CTMC approximation methods. For instance, Cui, Kirkby, and Nguyen (2019) in [11] analyzed the properties of CTMC approximation with more detailed operator structures in the generator matrix, extending their earlier work in [9]. Meier, Li, and Zhang (2021) in [13] and [15] explored CTMC approximations for one- and multi-dimensional diffusions with sticky boundary or interior points, which are particularly useful for problems such as modeling short-rate dynamics in low-interest-rate environments. CTMC approximation methods have also been applied to American option pricing. For example, Ma, Yang, and Cui (2021) in [12] employed CTMC methods to approximate the underlying asset dynamics and derived a closed-form expression for the early exercise premium (EEP) term. Additionally, Kaneko (2024) in [16] proposed a framework for constructing backward stochastic differential equations (BSDEs) driven by CTMCs through spatial discretization in both one and higher dimensions. Moreover, Lei, Zhou, and Xiao (2024) in [17] used analytical double transform inversions under CTMC approximations for pricing Asian options. Meanwhile, Li and Zhang (2019) in [10] also conducted a discussion on grid design and convergence analysis of the CTMC approximation in derivative pricing based on the spectral decomposition of the operator, which is quite different from ours.

The analysis in this report is structured around three major components. First, we present a detailed derivation of the convergence conditions for the Markov chain approximation operator. Next, we discuss the trade-off criteria for selecting grid design methods and prove that it can achieve higher accuracy in specific circumstances. Finally, we propose an innovative algorithm for parameter estimation based on the CTMC approximation methodology and analyze the numerical results of its application.

This report is organized as follows: Chapter 2 introduces the preliminary concepts as providing the setup and assumptions for later proofs. In Chapter 3, we present the background on the diffusion operator. Chapter 4 defines the framework for CTMC approximation and derives its

convergence conditions for both diffusions without jumps and Lévy processes. Chapter 5 includes an error analysis for the approximate diffusion process without jumps and discusses the criteria for selecting the number of states and different state discretization methods. Chapter 6 presents an algorithm for Markov chain approximation to estimate parameters, along with examples for calibrating parameters and the numerical results from our study. The final Chapter concludes the report. Detailed proofs are provided in the Appendix.

Chapter 2

Preliminaries

Define a Markov process X_t on the state space $\mathbf{E}$. To ensure that the state space is well defined, we first discuss with the space.

Definition 2.1. A *Polish space* is a separable completely metrizable topological space; that is, a space homeomorphic to a complete metric space that has a countable dense subset.

Definition 2.2. A *continuous-time Markov chain* is a stochastic process $(X_t)_{t \geq 0}$ on $\mathbf{E}$ such that:

$$\mathbb{P}(X_{t+s} = j \mid X_t = i, X_k, k < t) = \mathbb{P}(X_{t+s} = j \mid X_t = i),$$

for all $i, j \in \mathbf{E}$, $t \geq 0$, and $s \geq 0$.

Above properties are so called Markov property, where the conditional probability that calculate the change in different states is defined as transition probability.

Definition 2.3. *Transition probability* is defined as $p(t, x, y)$, where t is transition time, x is the starting point and $y \in \mathbf{E}$ is the arrival set.

Remark 2.4. In fact, a Markov chain is uniquely determined by a family of transition probability.

Definition 2.5. *Markov transition kernel* P_t (or Markov operator), defined from transition probability by convolution

$$P_t f = \int p(t, x, dy) f(y)$$

where $f \in C_0(E)$ and $C_0(E)$ represents the set of all continuous real functions on $\mathbf{E}$ that tend to 0 at infinity.

Definition 2.6. A *transition semigroup* is the collection of transition kernels $(P_t)_{t \geq 0}$ if and only if satisfies following three properties.

If f is an indicator function, then $P_t f = P_t(x, y)$ indicate the transition probability.

(i) For every $x \in \mathbf{E}$, $P_0(x, dy) = \delta_x(dy)$.

(ii) For every $s, t \geq 0$ and $A \in \mathbf{E}$,

$$P_{t+s}(x, A) = \int_E P_t(x, dy) P_s(y, A)$$

(Chapman-Kolmogorov identity).

(iii) For every $A \in \mathbf{E}$, the function $(t, x) \mapsto P_t(x, A)$ is measurable with respect to the σ -field $\mathcal{B}(\mathbb{R}_+) \otimes \mathbf{E}$.

Remark 2.7. A semigroup $(T(t))$ on B is said to be strongly continuous if $\|T(t)\| < Me^{at}$ for every T on B .

For continuous Markov chain, the transition probability satisfies above Chapman-Kolmogorov identity that $P(s+t) = P(s)P(t) = P(t)P(s)$. When $s \rightarrow 0$, Roughly, this can be regarded as taking derivative on both side of C-K equation and get $P'(t) = QP(t) = P(t)Q$. Therefore, the infinitesimal generating operator Q can be solved out, but strict proof will not be discussed here.

Definition 2.8. The transitional character of a continuous-time Markov chain can be described by its rate matrix Q .

$$P(\Delta t) = e^{Q\Delta t} = \sum_{k=0}^{\infty} \frac{(Q\Delta t)^k}{k!}, \quad \Delta t > 0$$

Definition 2.9. Let $(P_t)_{t \geq 0}$ be a transition semigroup on $\mathbf{E}$. $(P_t)_{t \geq 0}$ is a **Feller semigroup** if and only if:

- (i) $\forall f \in C_0(E), P_t f \in C_0(E)$;
- (ii) $\forall f \in C_0(E), \|P_t f - f\| \rightarrow 0$ as $t \rightarrow 0$.

A stochastic process is a **Feller process** if its semigroup is Feller. The Feller process can guarantee good properties such as strong Markov property.

Definition 2.10. Set

$$D(\mathcal{G}) = \left\{ f \in C_0(E) : \frac{P_t f - f}{t} \text{ converges in } C_0(E) \text{ when } t \downarrow 0 \right\}$$

and, for every $f \in D(\mathcal{G})$,

$$\mathcal{G}f = \lim_{t \downarrow 0} \frac{P_t f - f}{t}.$$

Then $D(\mathcal{G})$ is a linear subspace of $C_0(E)$ and $\mathcal{G} : D(\mathcal{G}) \rightarrow C_0(E)$ is a linear operator called the infinitesimal **generator** of the semigroup $(P_t)_{t \geq 0}$. The subspace $D(\mathcal{G})$ is called the domain of $\mathcal{G}$.

Remark 2.11. A semigroup $(P_t)_{t \geq 0}$ is uniquely determined by its generator $\mathcal{G}$ (also including the domain $D(\mathcal{G})$).

Definition 2.12. Let $\lambda > 0$. The λ -**resolvent** of the transition semigroup $(P_t)_{t \geq 0}$ is the linear operator $R_\lambda := B(E) \rightarrow B(E)$ defined by

$$R_\lambda f(x) = \int_0^\infty e^{-\lambda t} P_t f(x) dt$$

for $f \in B(E)$ and $x \in E$.

So far, all the pre-knowledge relevant to this report has been presented, and contents such as 'Feller process' and so on can be used as hypotheses to prove in the following sections. Now let's go back to our inquiry.

Chapter 3

Background

3.1 Heat Equation

The origins of the diffusion equation can be traced back to the study of heat conduction. Now let us consider a simple example

$$u_t = \mu u_x + \sigma u_{xx}$$

Dimension analysis (without considering u_{xx}):

$$\frac{[u]}{[t]} = [\mu] \frac{[u]}{[x]} \Rightarrow [\mu] = \frac{[x]}{[t]}$$

Therefore, μ represents velocity; this is also the characteristic line in the linear transport equation. By considering change of variable, we can solve the equation with solution

$$u(x, t) = \frac{1}{\sqrt{4\pi\sigma t}} \int_{\mathbb{R}} e^{-\frac{(x+\mu t-y)^2}{4\sigma t}} g(y) dy$$

The solution is just a Brownian motion with a drift term μ and variance σ . This operator for heat diffusion is

$$\mathcal{L} = \mu \frac{\partial}{\partial x} + \sigma \frac{\partial^2}{\partial x^2}$$

In fact, the heat equation can be thought of as a special case of the diffusion equation, where the diffusing quantity is thermal energy.

3.2 Operator of Diffusion Processes

Now consider a time-homogeneous diffusion process X_t , which satisfies the following stochastic differential equation:

$$dX_t = \mu(X_t)dt + \sigma(X_t)dW_t \tag{3.1}$$

where W_t is a standard Brownian motion, and μ, σ are satisfying appropriate regularity conditions to make the equation have unique solution. If X_t follows a Feller process, the family of operators semigroup P_t is determined by its infinitesimal generator $\mathcal{G}$, where $\mathcal{G}$ satisfies

$$\mathcal{G}f(x) = \lim_{t \downarrow 0} \frac{P_t f - f}{t} \tag{3.2}$$

then for the diffusion given in (3.1), we have

$$\mathcal{G}f(x) = \mu(x)\frac{\partial f}{\partial x} + \frac{1}{2}\sigma^2(x)\frac{\partial^2 f}{\partial x^2} \quad (3.3)$$

where f is test functions.

Proof. By Itô formula, consider a arbitrarily differentiable function $f(x)$

$$df(X_t) = \mu(X_t)\frac{\partial f}{\partial x}dt + \sigma(X_t)\frac{\partial f}{\partial x}dW_t + \frac{1}{2}\sigma^2(X_t)\frac{\partial^2 f}{\partial x^2}dt \quad (3.4)$$

Take expectation on both sides, we have

$$\mathbb{E}[df(X_t)] = \left(\mathbb{E}[\mu(X_t)\frac{\partial f}{\partial x}] + \frac{1}{2}\mathbb{E}[\sigma^2(X_t)\frac{\partial^2 f}{\partial x^2}] \right) dt \quad (3.5)$$

then

$$\frac{\partial f}{\partial t} = \mu(x, t)\frac{\partial f}{\partial x} + \frac{1}{2}\sigma^2(x, t)\frac{\partial^2 f}{\partial x^2} \quad (3.6)$$

□

Hence, by Hille–Yosida theorem, the diffusion semigroup satisfies the condition $P_t = e^{t\mathcal{G}}$ is defined as **Feynman-Kac semigroup**. More concretely, when P_t acts on a function f and describes the evolution of the distribution to the partial differential equation in (3.6) over time t . The semigroup can be interpreted probabilistically as:

$$P_t f = \mathbb{E}_x[f(X_t)] = \int_{\mathbb{R}} f(x)p(x, t)dx$$

we can reformulate above equation as:

$$\mathbb{E}[df(X_t)] = \int_{\mathbb{R}} f(x)\frac{\partial p(x, t)}{\partial t}dx \quad (3.7)$$

$$= \int_{\mathbb{R}} \left(\mu(x, t)\frac{\partial f}{\partial x} + \frac{1}{2}\sigma^2(x, t)\frac{\partial^2 f}{\partial x^2} \right) p(x, t)dx \quad (3.8)$$

$$= \int_{\mathbb{R}} f(x) \left(\frac{\partial^2}{\partial x^2} \left(\frac{1}{2}\sigma^2(x, t)p(x, t) \right) - \frac{\partial}{\partial x} (\mu(x, t)p(x, t)) \right) dx \quad (3.9)$$

where we can get

$$\frac{\partial p(x, t)}{\partial t} = -\frac{\partial}{\partial x} (\mu(x, t)p(x, t)) + \frac{\partial^2}{\partial x^2} \left(\frac{1}{2}\sigma^2(x, t)p(x, t) \right) \quad (3.10)$$

This is the **Fokker-Planck equation**, since it is far beyond our topic, we're not going to talk too much about it.

As mentioned in the Preliminaries, Markov chains are also characterize by their infinitesimal generating operator, which is actually the intensity rate matrix or so called the Q matrix. In this report, we aim to find a generalized criteria for Q that best approximates the diffusion operator, balancing accuracy with the number of states.

Chapter 4

Methodology

First consider a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, and let a continuous stochastic process X_t follows the simple diffusion process in (3.1). We define X_t^h as a continuous-time Markov chain (CTMC) with a finite state space Π .

$$\Pi_n := \{x_i^h \mid -\infty \leq x_0^h \leq x_1^h \leq \dots \leq x_n^h \leq \infty\}$$

where the distance between each grid can be different. Without considering a jump term in the diffusion process, it is clear that the continuous-time Markov chain must follows a Birth & Death Process with a standard infinitesimal generating operator.

$$Q = \begin{pmatrix} -\lambda_0 & \lambda_0 & 0 & 0 & \dots \\ \mu_1 & -(\lambda_1 + \mu_1) & \lambda_1 & 0 & \dots \\ 0 & \mu_2 & -(\lambda_2 + \mu_2) & \lambda_2 & \dots \\ 0 & 0 & \mu_3 & -(\lambda_3 + \mu_3) & \dots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

Hence, once the Q matrix converge to the diffusion operator $\mu(x)\frac{\partial f}{\partial x} + \frac{1}{2}\sigma^2(x)\frac{\partial^2 f}{\partial x^2}$, the X_t^h weakly converge to X_t , we can indeed approximate the diffusion process by a continuous-time Markov chain.

Piccioni (1987) in [2] first come up with the CTMC approximation formula for Q matrix with an uniform grid, that is $x_i - x_{i-1} = h$.

$$Q = \begin{cases} q_{i,i-1} = \frac{1}{2h^2}\sigma^2(x_i^h) + \frac{1}{h}\mu_-(x_i^h), \\ q_{i,i+1} = \frac{1}{2h^2}\sigma^2(x_i^h) + \frac{1}{h}\mu_+(x_i^h), \\ q_{i,i} = -\frac{1}{h^2}\sigma^2(x_i^h) - \frac{1}{h}|\mu(x_i^h)|, \\ q_{i,j} = 0 \quad \forall j \neq i, i-1, i+1 \end{cases} \quad (4.1)$$

where $\mu(\cdot)_\pm = \max\{\pm\mu, 0\}$ and $|\cdot|$ means the absolute value.

It is easy to show this method converge in first moment. However, the second moment convergent are not satisfied for finite grids.

Proof.

$$\mathbb{E} (X_{t+\Delta t}^h - X_t^h) = (q_{i,i+1}h - q_{i,i-1}h) \Delta t \quad (4.2)$$

$$= \left(\frac{1}{2h^2} \sigma^2(x_i^h) + \frac{1}{h} \mu_+(x_i^h) - \frac{1}{2h^2} \sigma^2(x_i^h) + \frac{1}{h} \mu_-(x_i^h) \right) h \Delta t \quad (4.3)$$

$$= \mu(x_i^h) \Delta t \quad (4.4)$$

$$\mathbb{E} (X_{t+\Delta t}^h - X_t^h)^2 = (q_{i,i+1}h^2 + q_{i,i-1}h^2) \Delta t \quad (4.5)$$

$$= \left(\frac{1}{2h^2} \sigma^2(x_i^h) + \frac{1}{h} \mu_+(x_i^h) + \frac{1}{2h^2} \sigma^2(x_i^h) + \frac{1}{h} \mu_-(x_i^h) \right) h^2 \Delta t \quad (4.6)$$

$$= (\sigma^2(x_i^h) + h|\mu(x_i^h)|) \Delta t \quad (4.7)$$

$$\neq \sigma^2(x_i^h) \Delta t \text{ when } h \rightarrow 0 \quad (4.8)$$

□

A special solution for second moment convergence in coarsen grids are provided in Chourdakis (2004) in [6] by considering a symmetric form of q . They set $\mu_+(x_i^h) = -\mu_-(x_i^h)$ to ensure the second order convergence and subject to a necessary condition of spacing $h \leq \frac{\sigma^2(x_i^h)}{|\mu(x_i^h)|}$.

4.1 Convergent Condition Set up

Now, inspired by previous set up, it is time to discuss a general frame work for CTMC approximation convergent conditions in general. We first consider the form below with non-uniform grids:

$$Q = \begin{cases} q_{i,i+1} := \frac{1}{x_{i+1}^h - x_i^h} w_n(x_i^h) \\ q_{i,i-1} := \frac{1}{x_i^h - x_{i-1}^h} v_n(x_i^h) \\ q_{i,i} := - \left(\frac{1}{x_{i+1}^h - x_i^h} w_n(x_i^h) + \frac{1}{x_i^h - x_{i-1}^h} v_n(x_i^h) \right) \\ q_{i,j} = 0 \quad \forall j \neq i, i-1, i+1 \end{cases} \quad (4.9)$$

where w_n and v_n are arbitrary functions. It is nature to consider the intensity q in this way because when $x_{i+1}^h \rightarrow x_i^h$, q tend to infinity, which imply the states change happens at every time second, it is exactly a stochastic process.

Consider the operator Q act on any test functions f

$$Qf(x_i^h) = \sum_i (f(x_{i-1}^h)q_{i,i-1} + f(x_{i+1}^h)q_{i,i+1} + f(x_i^h)q_{i,i}) \quad (4.10)$$

$$= \sum_i ([f(x_{i-1}^h) - f(x_i^h)] q_{i,i-1} + [f(x_{i+1}^h) - f(x_i^h)] q_{i,i+1}) \quad (4.11)$$

$$= \sum_i \left([f(x_{i-1}^h) - f(x_i^h)] \frac{1}{x_i^h - x_{i-1}^h} v_n(x_i^h) + [f(x_{i+1}^h) - f(x_i^h)] \frac{1}{x_{i+1}^h - x_i^h} w_n(x_i^h) \right) \quad (4.12)$$

$$= \sum_i \left(- \left[f'(x_i^h) + \frac{1}{2} f''(x_i^h)(x_{i-1}^h - x_i^h) + o(x_{i-1}^h - x_i^h) \right] v_n(x_i^h) \right. \quad (4.13)$$

$$\left. + \left[f'(x_i^h) + \frac{1}{2} f''(x_i^h)(x_{i+1}^h - x_i^h) + o(x_{i+1}^h - x_i^h) \right] w_n(x_i^h) \right) \quad (4.14)$$

$$= \sum_i (f'(x_i^h)(w_n(x_i^h) - v_n(x_i^h)) \quad (4.15)$$

$$+ \frac{1}{2} f''(x_i^h) [(x_{i+1}^h - x_i^h)w_n(x_i^h) - (x_{i-1}^h - x_i^h)v_n(x_i^h) + o(x_{i+1}^h - x_{i-1}^h)]) \quad (4.16)$$

If we set this operator that converge to the diffusion operator

$$\sum_i \left(f'(x_i^h) \mu(x_i^h) + \frac{1}{2} f''(x_i^h) \sigma^2(x_i^h) \right)$$

we need the two conditions:

$$\lim_{n \rightarrow \infty} (w_n(x_i^h) - v_n(x_i^h)) = \mu(x_i^h) \quad (4.17)$$

and

$$\lim_{n \rightarrow \infty} ((x_{i+1}^h - x_i^h)w_n(x_i^h) - (x_{i-1}^h - x_i^h)v_n(x_i^h)) = \sigma^2(x_i^h) \quad (4.18)$$

Inspired from Piccioni formula, we could consider w_n and v_n contains two parts that one for μ and one for σ .

$$\begin{cases} w_n(x_i^h) = \delta_+(x_i^h) + \eta_+(x_i^h) \\ v_n(x_i^h) = \delta_-(x_i^h) + \eta_-(x_i^h) \end{cases} \quad (4.19)$$

Since we want δ control μ , let $\eta_+(x_i^h) = \eta_-(x_i^h)$, then

$$\delta_+(x_i^h) - \delta_-(x_i^h) = \mu(x_i^h) \quad (4.20)$$

Since we want η control σ^2 , we have

$$\lim_{n \rightarrow \infty} ((x_{i+1}^h - x_i^h)\delta_+(x_i^h) + (x_{i+1}^h - x_i^h)\eta_+(x_i^h) - (x_{i-1}^h - x_i^h)\delta_-(x_i^h) - (x_{i-1}^h - x_i^h)\eta_-(x_i^h)) = \sigma^2(x_i^h) \quad (4.21)$$

When $(x_{i+1}^h - x_i^h) \rightarrow 0$, it is clear that the below condition is satisfied. But if we want the second moment always being matched in coarsen grid, we need to find an additional condition to keep $(x_{i+1}^h - x_i^h)\delta_+(x_i^h) - (x_{i-1}^h - x_i^h)\delta_-(x_i^h) = 0$.

$$\lim_{n \rightarrow \infty} ((x_{i+1}^h - x_i^h)\eta(x_i^h) + (x_{i-1}^h - x_i^h)\eta(x_i^h)) = \sigma^2(x_i^h) \quad (4.22)$$

then η satisfies

$$\lim_{n \rightarrow \infty} \eta(x_i^h) = \frac{\sigma^2(x_i^h)}{(x_{i+1}^h - x_i^h) - (x_{i-1}^h - x_i^h)} \quad (4.23)$$

Therefore, we can always set a form with

$$\begin{cases} w_n(x_i^h) = \delta_+(x_i^h) + \frac{\sigma^2(x_i^h)}{(x_{i+1}^h - x_i^h) + (x_i^h - x_{i-1}^h)} \\ v_n(x_i^h) = \delta_-(x_i^h) + \frac{\sigma^2(x_i^h)}{(x_{i+1}^h - x_i^h) + (x_i^h - x_{i-1}^h)} \end{cases} \quad (4.24)$$

In (4.24), all the free variables rely on δ function, we only have to choose the correct δ . In summary, define $x_{i+1}^h - x_i^h := h_i$, above condition equation for choosing the δ function:

$$\begin{cases} h_i \delta_+(x_i^h) + h_{i-1} \delta_-(x_i^h) = 0 \\ \delta_+(x_i^h) - \delta_-(x_i^h) = \mu(x_i^h) \end{cases} \quad (4.25)$$

we can solve the equations to get δ function,

$$\begin{cases} \delta_+(x_i^h) = \frac{h_{i-1}\mu(x_i^h)}{h_i + h_{i-1}} \\ \delta_-(x_i^h) = -\frac{h_i\mu(x_i^h)}{h_i + h_{i-1}} \end{cases} \quad (4.26)$$

and w_n, v_n become

$$\begin{cases} w_n(x_i^h) = \frac{h_{i-1}\mu(x_i^h)}{h_i + h_{i-1}} + \frac{\sigma^2(x_i^h)}{h_i + h_{i-1}} \\ v_n(x_i^h) = -\frac{h_i\mu(x_i^h)}{h_i + h_{i-1}} + \frac{\sigma^2(x_i^h)}{h_i + h_{i-1}} \end{cases} \quad (4.27)$$

where we can get matrix Q

$$\begin{cases} q_{i,i+1} := \frac{h_{i-1}\mu(x_i^h) + \sigma^2(x_i^h)}{h_i(h_i + h_{i-1})} \\ q_{i,i-1} := \frac{-h_i\mu(x_i^h) + \sigma^2(x_i^h)}{h_{i-1}(h_i + h_{i-1})} \end{cases} \quad (4.28)$$

we can also formulate a condition matrix as:

$$\begin{pmatrix} 1 & 1 & 1 \\ -h_{i-1} & 0 & h_i \\ h_{i-1}^2 & 0 & h_i^2 \end{pmatrix} \begin{pmatrix} q_{i,i-1} \\ q_{i,i} \\ q_{i,i+1} \end{pmatrix} = \begin{pmatrix} 0 \\ \mu(x_i^h) \\ \sigma^2(x_i^h) \end{pmatrix} \quad (4.29)$$

Once the condition equation been satisfied, we can get our CTMC approximation methods in correct convergent form. Therefore, as long as the construction of the generating operator that satisfies the above matrix conditions is satisfied, when the degree of subdivision is sufficient, the Markov chain will eventually converge to a specific diffusion process under the operator.

4.2 Examples

According to Mijatovi and Pistorius (2009) in [7], a same result of Q matrix form are designed:

$$Q = \begin{cases} q_{i,i+1} := \frac{h_{i-1}\mu(x_i^h) + \sigma^2(x_i^h)}{h_i(h_i + h_{i-1})} \\ q_{i,i-1} := \frac{-h_i\mu(x_i^h) + \sigma^2(x_i^h)}{h_{i-1}(h_i + h_{i-1})} \\ q_{i,i} := -(q_{i,i-1} + q_{i,i+1}) \\ q_{i,j} = 0 \quad \forall j \neq i, i-1, i+1 \end{cases} \quad (4.30)$$

This is also the rate matrix Q we formalize in Chapter 4.1 equation (4.29).

Next, Lo and Skindilias (2014) in [8] developed an improved CTMC approximation method, which also satisfied the condition we proposed.

$$Q = \begin{cases} q_{i,i+1} := \left(\frac{1}{h_i}\mu_+(x_i^h) + \frac{\sigma^2(x_i^h) - (h_{i-1}\mu_-(x_i^h) + h_i\mu_+(x_i^h))}{h_i(h_i + h_{i-1})} \right) \\ q_{i,i-1} := \left(\frac{1}{h_{i-1}}\mu_-(x_i^h) + \frac{\sigma^2(x_i^h) - (h_{i-1}\mu_-(x_i^h) + h_i\mu_+(x_i^h))}{h_{i-1}(h_i + h_{i-1})} \right) \\ q_{i,i} := -(q_{i,i-1} + q_{i,i+1}) \\ q_{i,j} = 0 \quad \forall j \neq i, i-1, i+1 \end{cases} \quad (4.31)$$

where δ is defined as:

$$\frac{1}{h_i}\delta_+(x_i^h) = \frac{(h_i + h_{i-1})\mu_+(x_i^h) - (h_{i-1}\mu_-(x_i^h) + h_i\mu_+(x_i^h))}{h_i(h_i + h_{i-1})} \quad (4.32)$$

$$= \frac{h_{i-1}\mu_+(x_i^h) - h_{i-1}\mu_-(x_i^h)}{h_i(h_i + h_{i-1})} \quad (4.33)$$

$$= \frac{h_{i-1}(\mu_+(x_i^h) - \mu_-(x_i^h))}{h_i(h_i + h_{i-1})} \quad (4.34)$$

$$= \frac{h_{i-1}\mu(x_i^h)}{h_i(h_i + h_{i-1})} \quad (4.35)$$

then

$$\delta_+(x_i^h) = \frac{h_{i-1}\mu(x_i^h)}{h_i + h_{i-1}} \quad (4.36)$$

Similarly,

$$\delta_-(x_i^h) = -\frac{h_i\mu(x_i^h)}{h_i + h_{i-1}} \quad (4.37)$$

Obviously, condition(4.26) are satisfied, this meet the requirements of the CTMC approximation condition in equation (4.29).

4.3 Jump-Diffusion Process

For a jump-diffusion process, the chain must visit non-neighbouring states, which implies it is not a standard birth-death process. In this section, we will briefly discuss the conditions required to simulate the jump-diffusion process using MCA method. Let us first define a diffusion process with jump component as:

$$dX_t = \mu(X_t)dt + \sigma(X_t)dW_t + \int_{\mathbb{R}} h(X_t, y)N(dt, dy), \quad (4.38)$$

where $N(dt, dx)$ is the Poisson measure, $\nu(dy)$ is the Lévy measure, and $h(x, y)$ is the jump amplitude function. Therefore, we can approach our infinitesimal generator as

$$\mathcal{L}f(x) = \mu(x)\frac{\partial f}{\partial x} + \frac{1}{2}\sigma^2(x)\frac{\partial^2 f}{\partial x^2} + \int_{\mathbb{R}} [f(x + h(x, y)) - f(x) - h(x, y)f'(x)]\nu(dy) \quad (4.39)$$

Detailed proof left in the Appendix A.

We observe that the generator of a jump-diffusion process can be decomposed into two distinct components: a continuous operator associated with the diffusion term and a pure jump operator derived from the Lévy measure. Therefore, we can also construct a discrete jump generator Q^J in Markov chain approximation framework to approximate the effect of the Lévy measure $\nu(\cdot)$. The resulting stochastic process thus comprises two parts: a birth-death process (diffusive component) and a pure jump process. The generator of the continuous-time Markov chain can then be written as

$$Q = Q^C + Q^J$$

which can be used to approximate diffusion processes with jump components.

However, to ensure the convergence of the Markov chain approximation, the jump operator must satisfy certain conditions, particularly that the first and second instantaneous moments match. The expectation and variance generated by the jump component are

$$\mu_J(x_i) := \int_{\mathbb{R}} h(x_i, y)\nu(x_i, dy) \quad (4.40)$$

$$\sigma_J^2(x_i) := \int_{\mathbb{R}} h^2(x_i, y)\nu(x_i, dy) \quad (4.41)$$

Hence, a necessary condition for the jump component is

$$\int_{\mathbb{R}} h^2(x_i, y)\nu(dy) < \infty \quad (4.42)$$

which ensures that the variance exists.

Based on the technique of Mijatovi and Pistorius (2009) in [7], we obtain the following formulation. In a continuous-time Markov chain, the jump operator Q^J can be described as

$$Q_{i, i \cdot (1+y_i)}^J := \int_{m(y_{i-1})}^{m(y_i)} \nu(i, dy) \quad (4.43)$$

where $m(y_i)$ is defined as the midpoint of two adjacent state values (i.e. $\frac{x_i^h + x_{i-1}^h}{2}$). In their framework, a specific jump amplitude function is considered with $h(x, y) = xy$, which is commonly used in the lognorm-style asset price model. The corresponding discrete jump operator

is then given by:

$$Q_{ij}^J = \begin{cases} 0, & \text{if } i = 1 \text{ or } i = N, \\ q_{i,j} := \int_{m(y_{i-1})}^{m(y_i)} \nu(i, dy), & \text{if } i \neq 1, N \text{ and } i \neq j, \\ q_{i,i} := -\sum_{j \neq i}^N q_{i,j}, & \text{if } i \neq 1, N. \end{cases} \quad (4.44)$$

To generalize this construction, we consider an arbitrary jump amplitude function $h(x, y)$. This section we consider the transition from $x_i^h \rightarrow x_j^h$ for all $j \neq i \pm 1$. Then we have

$$x_j^h = x_i^h + h(x_i^h, y) \quad (4.45)$$

$$y := h^{-1}(x_i^h, x_j^h - x_i^h) \quad (4.46)$$

Accordingly, the discrete jump generator in our case becomes

$$Q_{x_i^h, x_i^h + h(x_i^h, y_i)}^J := \int_{h^{-1}(x_i^h, x_j^h - x_i^h)} \nu(x_i^h, dy) \quad (4.47)$$

so we then define the rate matrix Q as

$$Q_{ij}^J = \begin{cases} 0, & \text{if } i = 1 \text{ or } i = N, \\ q_{i,j}^J := \int_{h^{-1}(x_i^h, x_j^h - x_i^h)} \nu(i, dy), & \text{if } i \neq 1, N \text{ and } j \neq i \pm 1, \\ q_{i,i}^J := -\sum_{j \neq i} q_{i,j}^J, & \text{if } i \neq 1, N. \end{cases} \quad (4.48)$$

Similar to the analysis in last section, we consider the operator Q^J act on any test functions f

$$Q^J f(x_i^h) = \sum_i (f(x_j^h) q_{i,j}^J + f(x_i^h) q_{i,i}^J) \quad (4.49)$$

$$= \sum_i (f(x_j^h) - f(x_i^h)) q_{i,j}^J \quad (4.50)$$

$$= \sum_i \left(f'(x_i^h)(x_j^h - x_i^h) + \frac{1}{2} f''(x_i^h)(x_j^h - x_i^h)^2 + o((x_j^h - x_i^h)^2) \right) \cdot \int_{h^{-1}(x_i^h, x_j^h - x_i^h)} \nu(x_i^h, dy) \quad (4.51)$$

from which we deduce the convergence conditions

$$\lim_{n \rightarrow \infty} \sum_j (x_j^h - x_i^h) q_{i,j}^J = \int_{\mathbb{R}} h(x_i^h, y) \nu(x_i^h, dy) \quad (4.52)$$

$$\lim_{n \rightarrow \infty} \sum_j (x_j^h - x_i^h)^2 q_{i,j}^J = \int_{\mathbb{R}} h^2(x_i^h, y) \nu(x_i^h, dy) \quad (4.53)$$

It is clear that from the definition of $Q_{i,j}^J$, the above convergence holds. The right-hand side represents the expectation of the displacement arising from all jumps starting at x_i^h , with jump amplitude function h and jump intensity ν . The left-hand side represents the discrete counterpart, where we sum the jump amplitude function have the specific value $h(x_j^h, y) = x_j^h - x_i^h$ by their intensity, which also implies the expectation of jump displacement. This can be viewed as approximating the integral by partitioning it into different intervals and summing the jump size

times the jump rate — a standard Riemann approximation idea. We thus arrive at the final form of the full generator:

$$Q := Q^C + Q^J = \begin{cases} q_{i,i+1} := \left(\frac{1}{h_i} \mu_+(x_i^h) + \frac{\sigma^2(x_i^h) - (h_{i-1} \mu_-(x_i^h) + h_i \mu_+(x_i^h))}{h_i(h_i + h_{i-1})} \right) \\ q_{i,i-1} := \left(\frac{1}{h_{i-1}} \mu_-(x_i^h) + \frac{\sigma^2(x_i^h) - (h_{i-1} \mu_-(x_i^h) + h_i \mu_+(x_i^h))}{h_{i-1}(h_i + h_{i-1})} \right) \\ q_{i,j} = \int_{h^{-1}(x_i^h, x_j^h - x_i^h)} \nu(x_i^h, dy), \quad \forall j \neq i, i-1, i+1 \\ q_{i,i} := - \left(q_{i,i-1} + q_{i,i+1} + \sum_j q_{i,j} \right) \end{cases} \quad (4.54)$$

where the continuous term and jump term are independent.

While the convergence conditions for Markov chain approximations of jump-diffusion processes are well established, this framework can even be extended to approximate Lévy processes. However, in practice, handling the jump intensity function $h(x, y)$ remains challenging and the convergence for the Lévy process remains to be proved, and we will not further explore Lévy process approximation here.

4.4 Boundary Conditions

In this article, we do not focus on boundary value problems related to diffusion processes. However, we briefly explain the boundary conditions for a diffusion process. Consider a diffusion process with state space (L, U) , where $-\infty < L < U < \infty$. If the diffusion state space is an infinite interval, the two boundary points L and U are assumed to be inaccessible, which is called **Inaccessible Boundary**. Moreover, if the boundary points are finite, they are considered non-singular boundary points. A non-singular boundary point serves as both an entrance and an exit of boundary, and it can be further classified into:

- **Reflecting boundary:** The process is reflected back into the state space immediately at the boundary.
- **Absorbing boundary:** The process is absorbed at the boundary and cannot return to the state space.

A **sticky boundary** refers to a type of boundary condition where a process is not allowed to immediately escape back to the state space when it reaches the boundary, but rather, it "sticks" near the boundary for some time before eventually leaving. This behavior lies between a reflecting boundary and an absorbing boundary.

Remark 4.1. *In this report, we only consider one-dimensional diffusion and assume the boundary is finite, with $x_{\max}^h$ as the upper boundary and $x_{\min}^h$ as the lower boundary in the finite state space Π_n . In addition, the boundary points are regarded to be either reflecting or absorbing boundary during the construction of continuous-time Markov chain approximation.*

According to Meier, Li, and Zhang (2021) in [13], they states the sticky boundary diffusion can be defined as a weak solution to the specific stochastic differential equations:

$$dX_t = \mu(X_t) \mathbb{I}(X_t > l) dt + \sigma^2(X_t) \mathbb{I}(X_t > l) dW_t + \frac{1}{2} dL_t^l(X_t) \quad (4.55)$$

$$\mathbb{I}(X_t = l) dt = \frac{1}{2\rho} dL_t^l(X_t) \quad (4.56)$$

where $\rho \in (0, \infty)$ is positive representing the stickiness of X_t at boundary l , and L_t^l is a (right) local time process of X_t at boundary l , which defined as below:

$$L_t^s(X_t) := \lim_{\varepsilon \downarrow 0} \frac{1}{\varepsilon} \int_0^t \mathbb{I}(X_s \in [l, l + \varepsilon]) dX_s \quad (4.57)$$

It is important to observe that two additional types of boundary behavior can be derived from the sticky case by taking limits. For example, $\rho = 0$, it corresponds to an absorbing boundary, and $\rho = \infty$, it corresponds to a reflecting boundary. However, here we merely want to explain the assumptions about boundaries in this article and do not conduct a more in-depth discussion on sticky boundary diffusion problems.

Chapter 5

Error Analysis

5.1 Approximation error of semi-groups

Assume that the diffusion process X_t is a Feller process on the state space $\mathcal{E}$, with associated semigroup P_t and infinitesimal generator $\mathcal{L}$. Let $(X_t(n))_{n \in N}$ denotes a sequence of continuous-time Markov chain with state spaces $\mathbf{E}$, corresponding semigroups $P_t^{(n)}$ and generator matrices $Q^{(n)}$. For any $f \in D(\mathcal{L})$, where $D(\mathcal{L})$ is the domain of $\mathcal{L}$, consider the following error measure $\epsilon(\cdot)$:

$$\epsilon(f) = \mathbf{Sup} \|P_t f - P_t^{(n)} f\| \leq t \cdot \mathbf{Sup} \|\mathcal{L} f - Q^{(n)} f\| \quad (5.1)$$

This inequality follows Ethier and Kurtz (1986) in [18] Proposition 2.7

$$\|P_t f - P_t^{(n)} f\| \leq t \cdot \|\mathcal{L} f - Q^{(n)} f\|$$

for any $f \in C_0(\mathbb{E})$. In equation (4.13) and (4.14), we can find the error term is

$$o(x_{i+1}^h - x_i^h)w_n(x_i^h) - o(x_{i-1}^h - x_i^h)v_n(x_i^h) \quad (5.2)$$

If we put (4.27) into the equation, we may get

$$o(h_i) \left(\frac{h_{i-1}\mu(x_i^h)}{h_i + h_{i-1}} + \frac{\sigma^2(x_i^h)}{h_i + h_{i-1}} \right) + o(h_{i-1}) \left(-\frac{h_i\mu(x_i^h)}{h_i + h_{i-1}} + \frac{\sigma^2(x_i^h)}{h_i + h_{i-1}} \right) = o(\sigma^2(x_i^h)) \sim o(1) \quad (5.3)$$

This implies that continuous-time Markov chain with the generator $Q^{(n)}$ converges weakly to diffusion process generator $\mathcal{L}$, as N (number of states grids) goes to infinity. Furthermore, we

could use higher order Tayler expansion to find out the error term.

$$\|\mathcal{L}f - Q^{(n)}f\| = \sum_i \|\mu(x_i^h)f'(x_i^h) + \frac{1}{2}\sigma^2(x_i^h)f''(x_i^h) - (f(x_{i+1}^h)q_{i,i+1} + f(x_{i-1}^h)q_{i,i-1} + f(x_i^h)q_{i,i})\| \quad (5.4)$$

$$= \sum_i \|\mu(x_i^h)f'(x_i^h) + \frac{1}{2}\sigma^2(x_i^h)f''(x_i^h) - \left(\left[f'(x_i^h) + \frac{1}{2!}f''(x_i^h)(x_{i+1}^h - x_i^h) \right. \right. \quad (5.5)$$

$$\left. + \frac{1}{3!}f'''(x_i^h)(x_{i+1}^h - x_i^h)^2 + o(x_{i+1}^h - x_i^h)^2 \right] w_n(x_i^h) + (-1) \left[f'(x_i^h) \right. \quad (5.6)$$

$$\left. - \frac{1}{2!}f''(x_i^h)(x_i^h - x_{i-1}^h) + \frac{1}{3!}f'''(x_i^h)(x_i^h - x_{i-1}^h)^2 + o(x_i^h - x_{i-1}^h)^2 \right] v_n(x_i^h) \Big) \| \quad (5.7)$$

$$= \sum_i \left\| \frac{1}{3!}f'''(x_i^h) [(x_{i+1}^h - x_i^h)^2 w_n - (x_i^h - x_{i-1}^h)^2 v_n] \right. \quad (5.8)$$

$$\left. + o(x_{i+1}^h - x_i^h)^2 w_n - o(x_i^h - x_{i-1}^h)^2 v_n \right\| \quad (5.9)$$

$$\text{Let } h(n, x_i^h) := (x_{i+1}^h - x_i^h)^2 w_n(x_i^h) - (x_{i-1}^h - x_i^h)^2 v_n(x_i^h) \quad (5.10)$$

$$= \sum_i \left\| \frac{1}{3!}f'''(x_i^h)h(n, x_i^h) + o(h_{i+1}^2) \left(\frac{h_{i-1}\mu(x_i^h)}{h_i + h_{i-1}} + \frac{\sigma^2(x_i^h)}{h_i + h_{i-1}} \right) \right. \quad (5.11)$$

$$\left. - o(h_{i-1}^2) \left(-\frac{h_i\mu(x_i^h)}{h_i + h_{i-1}} + \frac{\sigma^2(x_i^h)}{h_i + h_{i-1}} \right) \right\| \quad (5.12)$$

$$= \sum_i \left\| \frac{1}{3!}f'''(x_i^h)h(n, x_i^h) + o(h_i h_{i-1} \mu(x_i^h) + (h_i - h_{i-1})\sigma^2(x_i^h)) \right\| \quad (5.13)$$

From above analysis, we can know that the error term is relate to the test function f and the grids design, which is the $h(n, x_i^h)$ function.

$$h(n, x_i^h) := h_i h_{i-1} \mu(x_i^h) + (h_i - h_{i-1}) \sigma^2(x_i^h) \quad (5.14)$$

In particular, when we consider uniform grids method, the error function become

$$|h(n, x_i^h)| = h^2 |\mu(x_i^h)| \quad (5.15)$$

It is clear that the convergent rate is mean square convergence, where error term equals to $o(h^2)$.

Assumption 5.1. For a sequence of grids Π_N , h is the distance between different grids.

$$\frac{\max_{x_i^h \in \Pi_N} (h_i)}{\min_{x_i^h \in \Pi_N} (h_i)} \leq C \quad (5.16)$$

This assumption suggests that both the maximum and minimum step sizes should decrease at a constant rates, which holds for all grid types typically used in practice.

Assumption 5.2. Through the technique of spectral decomposition, we can verify that the maximum step size must satisfy the following relationship with the coefficients.

$$\max_{x_i^h \in \Pi_N} (h_i) \leq \min_{x_i^h \in \Pi_N} \left(\frac{\sigma^2(x_i^h)}{|\mu(x_i^h)|} \right) \quad (5.17)$$

Under this assumption, we can conclude that for any differentiable function $f \in C_0(E)$, given the grids x_i^h , we can find that $f(x_i^h) \leq C_f$ is bounded without considering singularity. Therefore,

the error function is controlled by $h(n, x_i^h)$, since $\max h(n, x_i^h) \leq C \cdot \min h(n, x_i^h)$. This means that we only need to focus on the largest distance h when analysing the error. As a result, the error is determined only by the grid design $h(n, x_i^h)$.

$$\epsilon(f) = \|Pf - P_n f\| \leq t \|\mathcal{L}f - Q^{(n)}f\| \leq t \cdot C_{\text{constant}} \cdot \sum_i C_{f(x_i^h)} \cdot \max\{|h(n, x_i^h)|\} \quad (5.18)$$

Therefore, it is obvious to find that the error term is dominated by $e_h := \sum_i |h(n, x_i^h)|$. All we want to do is find the minimum value of e_h .

If the specific accuracy requirements and specific constants are given, we can design the outlaw points according to this accuracy. However, this is unachievable in practice, we have to make further discussion to present our criteria for choosing specific N under given accuracy in the future.

Remark 5.3. *Inequality (4.37) requires the semigroups to be strongly continuous on a Polish space, along with other assumptions. We will not discuss these conditions further here but will assume they are satisfied.*

Remark 5.4. *We can use different inequalities to calculate this error term. For instance, Proposition 3.7 in Mijatovic and Pistorius (2009) in [7] presents an alternative approach for generator error analysis. The idea is following:*

$$\epsilon(f) \leq C_f(s) h(n, x_i^h) \quad (5.19)$$

The proofs follows Ethier and Kurtz (1986) in [18] Lemma 6.2, and we show this proof in the Appendix B.

5.2 Grid Design Methods

A well-constructed grid should satisfy two essential criteria. First, it must provide adequate resolution in key regions, such as dense in the areas that most points distributed, to ensure an accurate Markov chain approximation of the state space. Second, it should span a sufficiently broad domain to effectively control truncation errors introduced by replacing an infinite state space with a finite one. Now let us focus on the first criteria.

Different discretisation methods for the state space can have different effects on the numerical results, below, we will introduce some popular discretisation methods that have been mentioned in the papers and explore the relationship with the error analysis.

- **Uniform Grids**

$$h = \frac{x_{max}^h - x_{min}^h}{N} \quad (5.20)$$

We could generate our state space Π_N by

$$x_i^h = x_0^h + ih \quad (5.21)$$

For non-uniform grids, we should consider what the data distributed, since stochastic differential equations are controlled by Brownian motion, and Brownian motion can be regarded as a Gaussian process, the points taken are mostly concentrated near the mean of the process. This means that we should choose a lattice method, which is denser at the mean (the points we give) and sparser at the boundaries. This is in line with our empirical judgment.

- **Case A Non-uniform grids——Tavella and Randall grids**

Tavella and Randall (2000) proposed a concentrate grid point method where spatial variations are more concentrated around the drift term.

Consider a transformation as

$$x^h(u) = x_{midpoint}^h + \alpha \sinh(c_1 u + c_2(1 - u)) \quad (5.22)$$

where $u \in (0, 1)$ is uniform distributed and α is a prescribed uniformity parameter, then we have

$$c_1 = \operatorname{arcsinh}\left(\frac{x_{max}^h - x_{midpoint}^h}{\alpha}\right), \quad c_2 = \operatorname{arcsinh}\left(\frac{x_{min}^h - x_{midpoint}^h}{\alpha}\right) \quad (5.23)$$

We could generate our state space Π_N by

$$x_i^h = x_{midpoint}^h + \alpha \sinh\left(c_1 \frac{i}{N} + c_2 \left(1 - \frac{i}{N}\right)\right), \quad i = 2, 3, \dots, N-1 \quad (5.24)$$

In fact, $x_{midpoint}^h$ can be the critical points that h_i is minimized around it. Therefore, for $|x_i - x_{critical}^h| \gg \alpha$, the transformation is approximately to logarithmic function; and for $|x_i - x_{critical}^h| \ll \alpha$, the transformation is nearly uniform.

- **Case B Non-uniform grids——Mijatovi and Pistorius grids**

Mijatovi and Pistorius (2009) in [7] presents one method that is a slight modification version of Tavella and Randall grid design.

$$x_i^h = \begin{cases} x_0^h + \sinh\left(\frac{2-i}{\frac{N}{2}-1} \operatorname{arcsinh}(x_1^h - x_0^h)\right), & i = 2, \dots, \frac{N}{2}, \\ x_0^h + \sinh\left(\frac{i-\frac{N}{2}}{\frac{N}{2}} \operatorname{arcsinh}(x_N^h - x_0^h)\right), & i = \frac{N}{2} + 1, \dots, N-1. \end{cases} \quad (5.25)$$

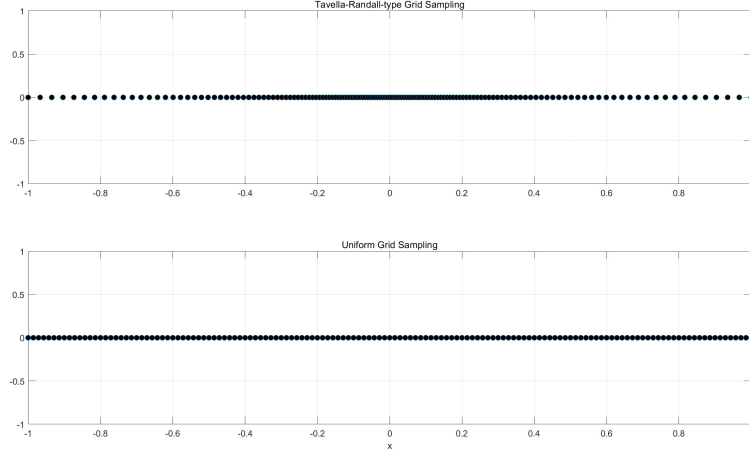


Figure 5.1: An Example of Uniform grids and Tavella-Randall grids

- **Case C Non-uniform grids—Lo and Skindilias grids**

Lo and Skindilias (2014) in [8] suggested a grid method based on the minimization of the Kolmogorov-Smirnov distance between the original cumulative distribution function (CDF) and the CTMC counterpart, where we define

$$x_t^h = F_{x_t^h}^{-1} \left(\frac{2i-1}{2N} \right) \quad (5.26)$$

and the theoretical distribution is defined as:

$$x_t^h \sim \frac{\chi^2 \left(\frac{4\eta\theta}{\sigma_{x^h}^2}, x_0^h c_t e^{-\eta t} \right)}{c_t} \quad (5.27)$$

where

$$c_t = \frac{4\eta}{\sigma_{x^h}^2 (1 - e^{-\eta t})} \quad (5.28)$$

Remark 5.5. Let F and G be two cumulative distribution functions. The **Kolmogorov-Smirnov distance** between F and G is defined as:

$$D_{KS}(F, G) = \sup_{x \in \mathbb{R}} |F(x) - G(x)| \quad (5.29)$$

- **Case D Non-uniform grids—Chebyshev grids**

The Chebyshev grids of the first kind on the interval $[-1, 1]$ are defined by:

$$x_k^h = \cos \left(\frac{2k-1}{2n} \pi \right), \quad k = 1, 2, \dots, n. \quad (5.30)$$

It is one of the most popular non-uniform grid method in numerical computing. However, it typically results a sequence of grids that is dense on both sides and sparse in the middle, which is contradict to our statement above. Hence, we will not discuss this here.

- **Case E Non-uniform grids—special case**

In this grids method, we consider different uniform densities in different intervals. This lattice discrete design is case by case, so here we only citation one example in other paper, and we will not further discuss the idea in this article. For example, Zhang and Li (2019) in [10] discussed this method for barrier option pricing consists of four parts:

1. $\Pi_1 = \{x_0^h + ih_1 \mid 0 \leq i \leq n_1\}$ with $h_1 = \frac{L-x_0}{n_1}$;
2. $\Pi_2 = \{L + \frac{h_2}{2} + ih_2 \mid 0 \leq i \leq n_2\}$ with $h_2 = \frac{K-L}{n_2}$;
3. $\Pi_3 = \{K + \frac{h_2}{2} + ih_3 \mid 0 \leq i \leq n_3\}$ with $h_3 = \frac{R-K-\frac{h_2}{2}}{n_3}$;
4. $\Pi_4 = \{R + ih_4 \mid 0 \leq i \leq n_4\}$ with $h_4 = \frac{x_N^h - R}{n_4}$.

where K is strike price, L and R is the barrier.

There are many different grid design methods, the analysis idea is the same as above. In this section, we focus on the Tavella and Randall grid design.

5.3 Non-uniform Grid High-Order Approximation

In the analysis of non-uniform grid design, we clarify that a non-uniform grid can be treated as a nonlinear transformation of a uniform grid. This idea is also reflected in formulas (5.18) and (5.19). To this end, we define all non-uniform grid points as functions of the uniform grid points, as follows:

$$x_i^h = \varphi(u_i); \quad \varphi(u_i) = au_i + c \text{ is linear, it is also uniform grids} \quad (5.31)$$

where u_i represents the uniform grid points. The distance between consecutive grid points can then be defined as

$$h_i = x_{i+1}^h - x_i^h = \varphi(u_{i+1}) - \varphi(u_i) = \varphi'(u_i)\Delta u + \frac{1}{2}\varphi''(u_i)\Delta u^2 + o((\Delta u)^2) \quad (5.32)$$

Thus, we obtain the error function in equation (5.14) as

$$\sum_i h(n, x_i^h) = \sum_i h_i h_{i-1} \mu(x_i^h) + (h_i - h_{i-1})\sigma^2(x_i^h) \quad (5.33)$$

$$= \sum_i (\Delta u)^2 [\varphi'(u_i)^2 \mu(\varphi(u_i)) + \varphi''(u_i)\sigma^2(\varphi(u_i))] + o((\Delta u)^4) \quad (5.34)$$

Furthermore, if we consider the uniform grid design in (5.15), we know that $h = (x_{\max}^h - x_{\min}^h)\Delta u$, and $x_i^h = \varphi_{\text{linear}}(u_i) = (x_{\max}^h - x_{\min}^h)u_i + x_{\min}^h$. Changing the notation, the error function for uniform grids become

$$\sum_i h(n, x_i^h) = \sum_i (x_{\max}^h - x_{\min}^h)^2 \Delta u^2 \cdot |\mu(\varphi_{\text{linear}}(u_i))| \quad (5.35)$$

Comparing the uniform and non-uniform grids, we find a comparison between

$$\sum_i (x_{\max}^h - x_{\min}^h)^2 \Delta u^2 \cdot |\mu(\varphi_{\text{linear}}(u_i))| \text{ v.s. } \sum_i \Delta u^2 [\varphi'(u_i)^2 \mu(\varphi(u_i)) + \varphi''(u_i)\sigma^2(\varphi(u_i))] \quad (5.36)$$

Hence, once the function φ satisfies the equation $E_n > 0$:

$$E_n := \sum_i \left(|\mu(\varphi_{\text{linear}}(u_i))| - \frac{1}{(x_{\max}^h - x_{\min}^h)^2} |\varphi'(u_i)^2 \mu(\varphi(u_i)) + \varphi''(u_i)\sigma^2(\varphi(u_i))| \right) \quad (5.37)$$

we can conclude that non-uniform grids can indeed reduce the error and improve precision.

For a special case, we consider μ and σ^2 are constant, then the E_n function become:

$$\sum_i \left(|\mu| - \frac{1}{(x_{\max}^h - x_{\min}^h)^2} |\varphi'(u_i)^2 \mu + \varphi''(u_i)\sigma^2| \right) \quad (5.38)$$

Now, let's focus on the Tavella and Randall grid design and define it as

$$x_i^h = \varphi(u_i) = x_{\text{midpoint}}^h + \alpha \sinh(c_1 u_i + c_2(1 - u_i)) \quad (5.39)$$

Alternatively, this can also be written as

$$x_i^h = \varphi(u_i) = x_{\text{midpoint}}^h + \alpha \sinh(c_2 + (c_1 - c_2)u_i) \quad (5.40)$$

where c_1 and c_2 are defined as:

$$c_1 = \operatorname{arcsinh}\left(\frac{x_{\max}^h - x_{\text{midpoint}}^h}{\alpha}\right), \quad c_2 = \operatorname{arcsinh}\left(\frac{x_{\min}^h - x_{\text{midpoint}}^h}{\alpha}\right) \quad (5.41)$$

Through these definitions, we can construct a non-uniform grid design analysis. Next, we can calculate the first derivative of this grid, which is given by:

$$\varphi'(u_i) = \alpha(c_1 - c_2) \cosh(c_2 + (c_1 - c_2)u_i) \quad (5.42)$$

Similarly, the second derivative is given by:

$$\varphi''(u_i) = \alpha(c_1 - c_2)^2 \sinh(c_2 + (c_1 - c_2)u_i) \quad (5.43)$$

Furthermore, we can compute $c_1 - c_2$ in advance. Since

$$x_{\max}^h - x_{\text{midpoint}}^h = 2(x_{\max}^h - x_{\min}^h), \quad x_{\min}^h - x_{\text{midpoint}}^h = -2(x_{\max}^h - x_{\min}^h) \quad (5.44)$$

we know that $c_1 = -c_2$, we can also define $m := x_{\max}^h - x_{\min}^h$ and $\xi := \frac{m}{\alpha}$, then

$$c_1 - c_2 = 2 \operatorname{arcsinh}\left(\frac{\xi}{2}\right) = 2 \ln\left(\frac{\xi}{2} + \sqrt{\left(\frac{\xi}{2}\right)^2 + 1}\right) \quad (5.45)$$

Lemma 5.6.

$$\operatorname{arcsinh}(x) = \ln\left(x + \sqrt{x^2 + 1}\right) \quad (5.46)$$

Then, let us discuss with the error function, plug in equation (5.39) and (5.40) into (5.35), it becomes

$$E_n = \sum_i \left(|\mu_i| - \left| \frac{1}{\xi^2} (c_1 - c_2)^2 \cosh^2(c_2 + (c_1 - c_2)u_i) \mu_i + \frac{1}{m} \frac{1}{\xi} (c_1 - c_2)^2 \sinh(c_2 + (c_1 - c_2)u_i) \sigma_i^2 \right| \right) \quad (5.47)$$

$$= \sum_i \left(|\mu_i| - \left| \frac{1}{\xi^2} \left(2 \ln\left(\frac{\xi}{2} + \sqrt{\left(\frac{\xi}{2}\right)^2 + 1} \right) \right)^2 \cosh^2(c_2 + (c_1 - c_2)u_i) \mu_i \right. \right. \quad (5.48)$$

$$\left. + \frac{1}{m} \frac{1}{\xi} \left(2 \ln\left(\frac{\xi}{2} + \sqrt{\left(\frac{\xi}{2}\right)^2 + 1} \right) \right)^2 \sinh(c_2 + (c_1 - c_2)u_i) \sigma_i^2 \right| \right) \quad (5.49)$$

$$\sim \sum_i \left(|\mu_i| - \left| \frac{1}{\xi^2} 4 \ln^2(\xi) \cosh^2(c_2 + (c_1 - c_2)u_i) \mu_i + \frac{1}{m} \frac{1}{\xi} 4 \ln^2(\xi) \sinh(c_2 + (c_1 - c_2)u_i) \sigma_i^2 \right| \right) \quad (5.50)$$

Then the key point is to analysis $\cosh(c_2 + (c_1 - c_2)u_i)$ and $\sinh(c_2 + (c_1 - c_2)u_i)$ where $u_i \in (0, 1)$. From our above analysis, we can define $\gamma_i := (c_2 + (c_1 - c_2)u_i) = (2u_i - 1) \operatorname{arcsinh}\left(\frac{\xi}{2}\right)$.

Through Lemma 5.6 we know that: $\gamma_i \sim (2u_i - 1) \ln(\xi)$. Therefore, we have:

$$\cosh(\gamma_i) \sim \frac{1}{2} \xi^{|2u_i - 1|}, \quad \sinh(\gamma_i) \sim \frac{1}{2} \xi^{|2u_i - 1|} \cdot \operatorname{sign}(2u_i - 1) \quad (5.51)$$

where $\operatorname{sign}(x)$ is the sign function. Since u_i is symmetric about point 0.5 on interval $(0, 1)$, we know that the second term in error function $\frac{1}{m} \frac{1}{\xi} 4 \ln^2(\xi) \sinh(c_2 + (c_1 - c_2)u_i) \sigma_i^2$ will be zero when the σ is a constant. This is due to the symmetry of the uniform grid points on $(0, 1)$ interval, and it simplifies the error function considerably. Moreover, the full error function can be expressed as:

$$E_n \sim \sum_i \left(|\mu_i| - \left| 2 \ln^2(\xi) \xi^{2|2u_i - 1| - 2} \mu_i + \frac{1}{m} 2 \ln^2(\xi) \xi^{|2u_i - 1| - 1} \cdot \operatorname{sign}(2u_i - 1) \sigma_i^2 \right| \right) \quad (5.52)$$

This expression gives us a formula to evaluate the operator error for the Tavella and Randall grid design. In numerical analysis, ξ is usually a bounded value, but it belongs to $(0, \infty)$ (i.e. The

common value is around 10,000). While this formula provides a relatively flexible expression, the values at different grid points will vary. Therefore, this formula primarily serves as a guideline for numerical selection.

It is important to note that this formula implies that non-uniform grid points design are not universally applicable to all problems. Depending on the value of ξ , we may obtain different results. The effectiveness of non-uniform grids largely depends on the characteristics of the problem at hand, especially how the parameter distribution changes. In cases where the parameter distribution exhibits clear, identifiable changes at specific positions, non-uniform grids can potentially reduce the error and improve accuracy. However, for problems where the parameters remain constant or do not exhibit such variation, non-uniform grids might introduce unnecessary errors and lead to a reduction in accuracy.

Based on our analysis, when the parameters are constant, the uniform grid points tend to be more effective. In this case, using uniform grids avoids the additional complexity introduced by non-uniform grids and better captures the underlying characteristics of the data. Non-uniform grid points, on the other hand, are more beneficial when the parameters vary with the grid points, as they can adapt to changes in the parameter distribution. However, such cases require careful analysis. Only when the distribution of the parameter function demonstrates clear patterns at certain points can we confidently state that non-uniform grid point design will improve the accuracy of the calculation. Nevertheless, many aspects of the above discussion still require further examination, as the choice between uniform and non-uniform grids is context-dependent and may vary with different problems and parameter behaviors.

Chapter 6

Numerical Implement in Finance

6.1 Markov Chain Approximation for Parameter Estimation

In most financial models, the drift and volatility terms of a process involve several unknown parameters that must be estimated before the model can be used. In a typical setting with constant parameters, there are several standard methods for parameter estimation, including moment estimation, maximum likelihood estimation, and Bayesian estimation. Now, consider a standard stochastic differential equation:

$$dX_t = \mu dt + \sigma^2 dW_t \quad (6.1)$$

At a specific time t , this can be treated as a normal distribution, so all of the aforementioned estimation methods can be applied to this case.

However, when the constant parameters become functions of time and state, such as $\mu(X_t, t)$ and $\sigma^2(X_t, t)$, the equation becomes

$$dX_t = \mu(X_t, t)dt + \sigma^2(X_t, t)dW_t \quad (6.2)$$

In this case, the parameters themselves evolve as a stochastic process. The key question then becomes: how can we estimate these dynamic parameters?

For example, in one of the most common financial model—stochastic volatility model, we have

$$\begin{cases} dS_t = \mu(v_t, S_t)dt + \sigma^2(v_t, S_t)dW_t \\ dv_t = \mu_0(v_t)dt + \sigma_0^2(v_t)dW_t \end{cases} \quad (6.3)$$

Here, the parameters μ and σ depend on a volatility process v_t , which is itself stochastic. As a result, following methods such as Markov Chain Monte Carlo (MCMC), the Kalman filter, or the particle filters are commonly used to estimate these parameter processes. A more detailed introduction with these methods are provided in Appendix C.

Since a continuous-time Markov chain (CTMC) approximation is capable of characterizing a stochastic process, it is naturally regarded as an effective tool for parameter processes estimation. Cui, Kirkby, and Nguyen (2017) in [9] combined this approach with a regime-switching

model. Specifically, in a stochastic volatility framework where parameters depend solely on the volatility process, they estimated the volatility using the CTMC approximation and assigned the resulting values as different parameter regimes in the stock price process.

Inspired by [9], we aim to refine the CTMC approximation as a more powerful method for parameter estimation, allowing us to directly estimate parameter processes in various complex financial models. For general diffusion processes, the transition operator plays a central role in governing the evolution of distributions. However, since the transition density is difficult to know for most diffusion processes, the CTMC approximation framework has the natural advantages and can become one of the most effective ways to implement semigroup-based parameter estimation. Additionally, comparing this method with other estimation techniques is also crucial for understanding the strengths and limitations of the CTMC approach.

6.2 Algorithm Set up

The key idea behind the algorithm is that the infinitesimal generator matrix Q theoretically contains all the information related to the diffusion process, and thus inherently contains the parameters Θ . Consequently, the primary challenge in CTMC approximation lies in accurately estimating the rate matrix Q from data that are discrete in time but continuous in the state space. While Chapter 3 outlined the CTMC approximation conditions that guarantee convergence, which yielding a theoretical form of rate matrix $Q(\Theta)$, we now combine this with grid construction in Chapter 4 to develop an algorithm suitable for numerical implementation.

In real-world financial problems, we often deal with time-series data $(Y_t)_{t \geq 0}$, such as $\{y_1, y_2, \dots, y_n\}$. Our approach so far mirrors the idea of the "Lebesgue integral", where we discretize the continuous state space into n intervals, with changing the whole state space $(x_{min}^h - \epsilon, x_{max}^h + \epsilon)$. The real data $(Y_t)_{t \geq 0}$ can then be mapped to corresponding state labels. For instance, a state sequence like $\{1, 1, 1, 2, 3, 3, 2, 1, \dots\}$ may represent the original observations $\{1.1, 1.0, 0.9, 2.2, 3.05, 2.99, \dots\}$. This process effectively transforms the real data into a Markov chain data on a discretized state space. Moreover, financial time series are typically recorded at uniform intervals Δt (e.g., daily observations), which implies that the data can naturally form a specific Markov chain path. Typically, a discrete finite sample path from a single process can serve to estimate unknown parameters. Using this approach, we can construct the rate matrix $Q(\Theta)$, and then obtain the transition probability matrix by computing the matrix exponential.

$$P(t) = e^{tQ} = \sum_{n=0}^{\infty} \frac{(Qt)^n}{n!} \quad (6.4)$$

Subsequently, we apply the maximum likelihood estimation (MLE) method. Given an initial guess Θ_0 of the parameters, we seek the parameter value that maximizes the likelihood of generating the observed Markov path. The maximum likelihood function is define as

$$l(\theta) = \sum_{i,j} \mathbf{log} \mathbb{P}(X_t^h = j \mid X_{t-1}^h = i; \Theta) \quad (6.5)$$

where the measure $\mathbb{P}$ is the the element in the transition probability matrix P , and the maximum likelihood estimator (MLE) is defined as the solution that maximizes the following constrained optimization problem:

$$\hat{\Theta} = \operatorname{argmax}_{\Theta} l(\theta) \quad (6.6)$$

Once the optimal parameter estimate is obtained using this method, we can proceed with error analysis and integrate the estimation framework into a wide range of diffusion-based financial models.

6.3 An Example for O-U process

In this section, we apply our algorithm to the Ornstein–Uhlenbeck process for parameter estimation.

$$dX_t = \theta(\mu - X_t)dt + \sigma\sqrt{X_t}dW_t \quad (6.7)$$

where $\mu(X_t, t) = \theta(\mu - X_t)$ and $\sigma(X_t, t) = \sigma\sqrt{X_t}$ are indeed stochastic process, respectively. This example is well-suited to our framework, as it corresponds to the Cox-Ingersoll-Ross (CIR) model widely used in finance. We use this relatively simple case to test the effectiveness of our algorithm. The flowchart of the algorithm is illustrated below:

Algorithm 1: Markov Chain Approximation

Input: a real-world dataset Y with initial assumption parameter

Output: estimated parameters for the unobserved diffusion process

- 1 **Simulate O-U Process:** Generate a trajectory using:

$$x_{i+1} = x_i + \theta(\mu - x_i)\Delta t + \sigma\sqrt{x_i}\sqrt{\Delta t} \cdot \mathcal{N}(0, 1)$$

Then sample data along the time axis from this original process.

- 2 **Discretize State Space:** Partition original data range into grids space Π_N with interval $[x_{min}^h, x_{max}^h]$ and assign each point to a specific discrete state.
- 3 **Construct Theoretical Q Matrix:** We generate the theoretical rate matrix $Q(\Theta_0)$ under condition (4.29) with initial assumption parameter $\Theta_0 = (\theta_0, \sigma_0, \mu_0)$.

$$Q_{i,i\pm 1} = \max\left(\frac{\sigma^2 x_i^h}{2h^2} \pm \frac{\theta(\mu - x_i^h)}{2h}, 0\right)$$

$$Q_{ii} = -(Q_{i,i-1} + Q_{i,i+1})$$

where the matrix Q is related to our set up in Chapter 3.

- 4 **Estimate Parameters (MLE):** Maximize log-likelihood of transition probabilities $P = e^{tQ}$.

$$\operatorname{argmax}_{\Theta} l(\theta) = \sum_{i \neq j} \log \mathbb{P}(x_t^h = j \mid x_{t-1}^h = i; \Theta)$$

where the probability measure $\mathbb{P}$ is from the transition matrix P and determined by the matrix $Q(\Theta)$.

6.4 Numerical Result

In this section, we conduct a numerical study of the above algorithm applied with a uniform grid design. We begin by defining the parameters: θ denotes the mean reversion speed, μ the long-term mean level, σ the volatility, and $num - states$ the number of states in the continuous-time Markov chain. As an illustrative example, we consider the standard Ornstein-Uhlenbeck (O-U) process with parameter set (θ, σ, μ) , (also known as the Vasicek model) to demonstrate the approximation idea of the algorithm, as shown in Figure 6.1 and Figure 6.2.

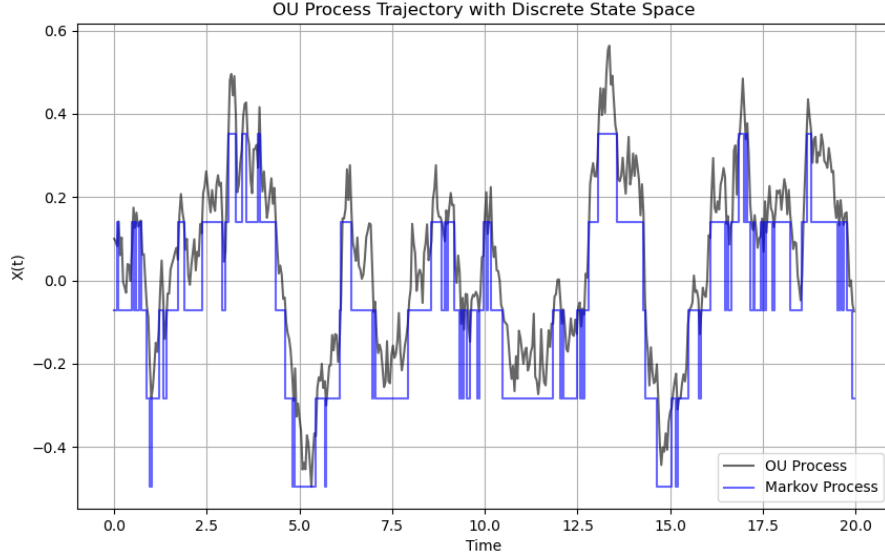


Figure 6.1: Algorithm implementation on Vasicek model with $(\theta = 1.0, \sigma = 0.3, \mu = 0, \text{number states} = 5)$

In order to enhance the applicability of our algorithm to real-world financial data, we revisit the simulation based on the CIR model and select parameter values that closely align with practical financial interpretations. At the same time, we engage in further discussions from a numerical computation perspective to optimize the algorithm.

Specifically, we assume the parameters to be $(\theta = 0.8, \sigma = 0.05, \mu = 0.03)$, which are broadly consistent with typical values observed in financial markets. Additionally, we set the total simulation time as $T = 50$ and the number of time steps as $N = 12500$, resulting in a time interval around $\frac{T}{N} = \frac{1}{252}$, a choice that mirrors the typical number of trading days in a year and thereby guides our selection of N .

Unless otherwise specified, all subsequent simulations in this section will adopt this parameter configuration, as other parameter choices have also yielded satisfactory results under the same numerical setup. Here, we treat this set of parameters as a representative case. Furthermore, the errors we calculate by default are all relative errors, which satisfy the formula:

$$\frac{\sqrt{\frac{1}{n} \sum_{i=1}^n (\Theta_i^{est} - \Theta^{real})^2}}{\Theta^{real}} \quad (6.8)$$

This formula eliminates the dimension and represents the signal-to-noise ratio of the information.

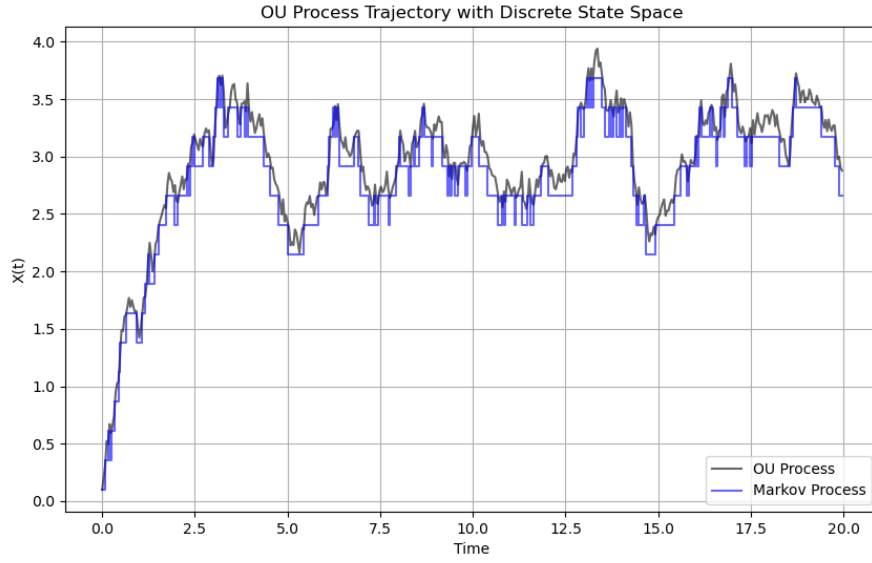


Figure 6.2: Algorithm implementation on Vasicek model with ($\theta = 1.0$, $\sigma = 0.5$, $\mu = 3.0$, number states = 15)

We now proceed to examine the error between the estimated parameters and the true parameter values. The results are presented in Figure 6.3 and Table 6.1 below:

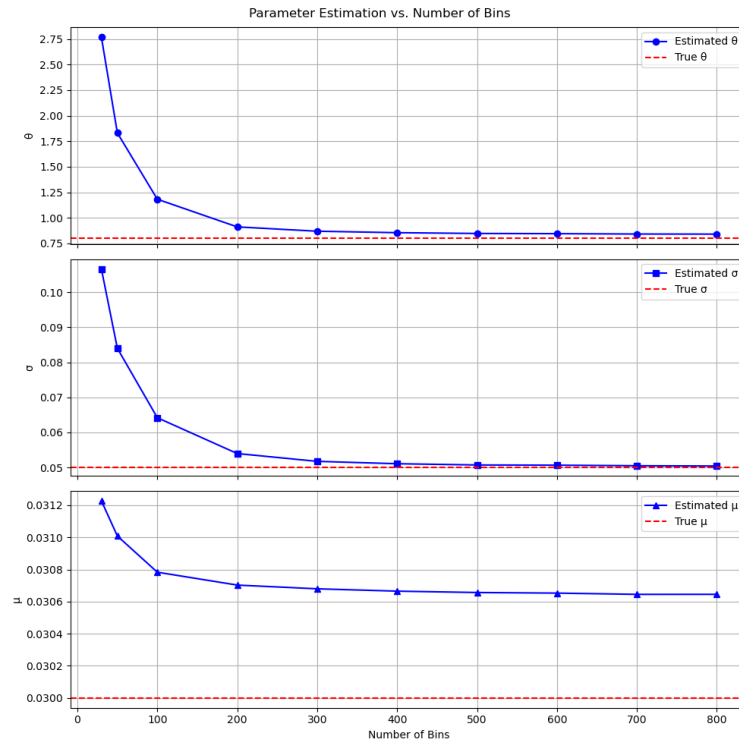


Figure 6.3: Parameter estimation error for CIR model under different discrete degrees of state Spaces

num_bins	θ	σ	μ	RelErr(θ)	RelErr(σ)	RelErr(μ)
30	2.7692	0.1067	0.0312	2.46150	1.13310	0.04092
50	1.8306	0.0840	0.0310	1.28826	0.68005	0.03361
100	1.1823	0.0642	0.0308	0.47790	0.28404	0.02610
200	0.9117	0.0540	0.0307	0.13964	0.07968	0.02343
300	0.8694	0.0518	0.0307	0.08677	0.03514	0.02265
400	0.8543	0.0511	0.0307	0.06792	0.02166	0.02231
500	0.8454	0.0507	0.0307	0.05669	0.01442	0.02196
600	0.8459	0.0507	0.0307	0.05734	0.01329	0.02174
700	0.8420	0.0505	0.0306	0.05255	0.01006	0.02150

Table 6.1: Estimates and Relative Error Table

From the above results, we observe that the estimated parameters tend to stabilize when the discretization level of the state space reaches approximately 200. This indicates a remarkably fast convergence, suggesting that a 200×200 rate matrix Q is already sufficient to capture the information embedded in the 12,500 data points.

However, the current data generation is based on a fixed random seed ($seed = 51$), which may introduce sampling bias. To reduce the influence of randomness, we further evaluate the parameter estimates by averaging the results across 5 more different random seeds. The corresponding graph and numerical results are presented in Figure 6.4 and Table 6.2:

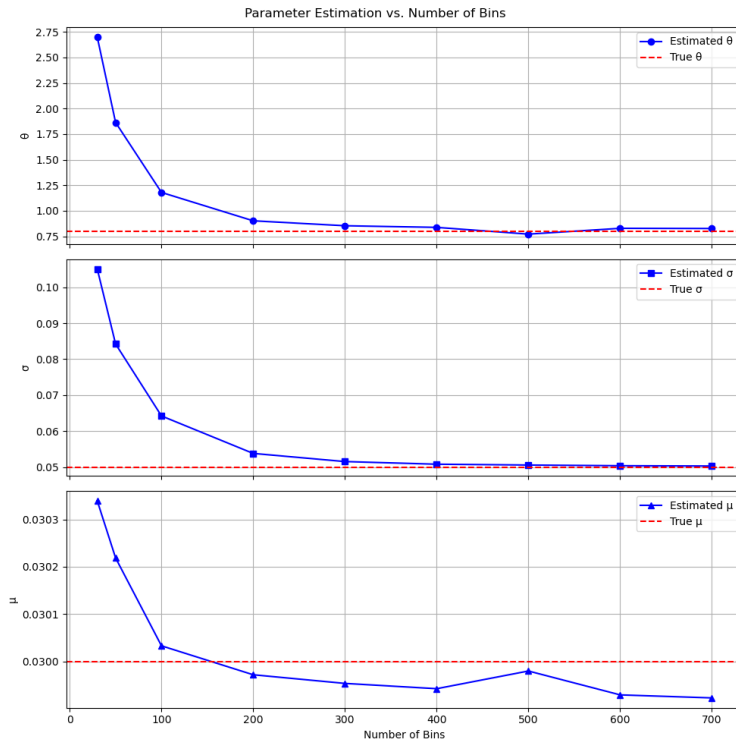


Figure 6.4: Average parameter estimation error for CIR model under different discrete degrees of state Spaces

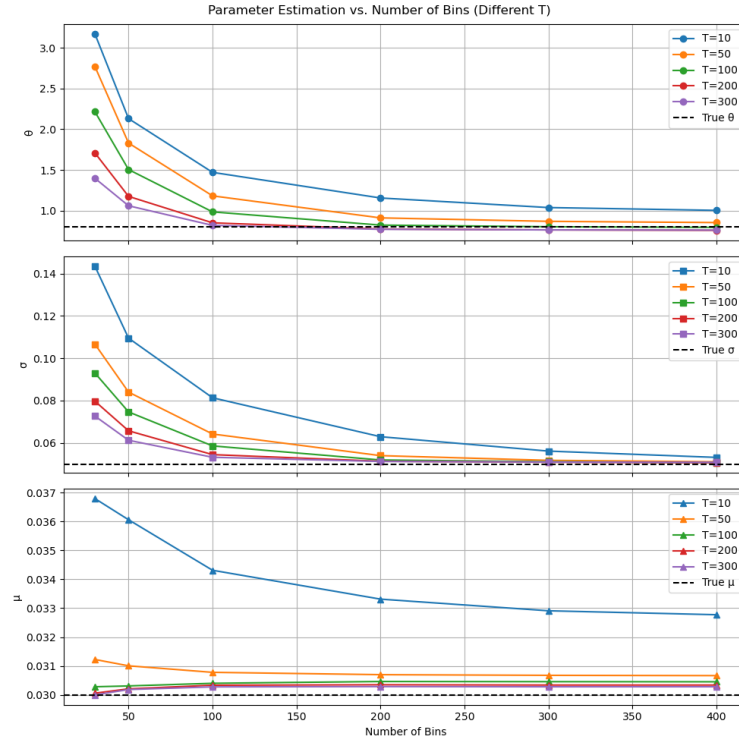
num_bins	θ (avg)	σ (avg)	μ (avg)	RelErr(θ)	RelErr(σ)	RelErr(μ)
30	2.6995	0.1051	0.0303	2.38102	1.10257	0.02539
50	1.8634	0.0842	0.0302	1.32997	0.68464	0.01949
100	1.1791	0.0642	0.0300	0.47544	0.28444	0.01919
200	0.9014	0.0538	0.0300	0.13347	0.07583	0.01849
300	0.8530	0.0515	0.0300	0.07794	0.03056	0.01870
400	0.8368	0.0508	0.0299	0.06100	0.01618	0.01853
500	0.7712	0.0506	0.0300	0.17903	0.01212	0.01923
600	0.8279	0.0504	0.0299	0.05290	0.00888	0.01878
700	0.8259	0.0503	0.0299	0.05147	0.00741	0.01863

Table 6.2: Average Estimates and Relative Error Table

Remark 6.1. Note that the spacing units of y-axis in the Figures are different.

These results demonstrate that the algorithm is highly efficient for parameter estimation, with errors falling within an acceptable range. In addition, we observe that the degree of time discretization is closely related to the accuracy of parameter estimation during the experimental process. This phenomenon is explored in the following.

Interestingly, we find that when the time subdivision is relatively coarse, the parameter fitting tends to be more accurate. To investigate this further, we conduct experiments by fixing T and N separately for parameter estimation. The corresponding results are illustrated in Figure 6.5 and Figure 6.6.

Figure 6.5: Fix $N = 12500$, set different $T = 10, 50, 100, 200, 300$

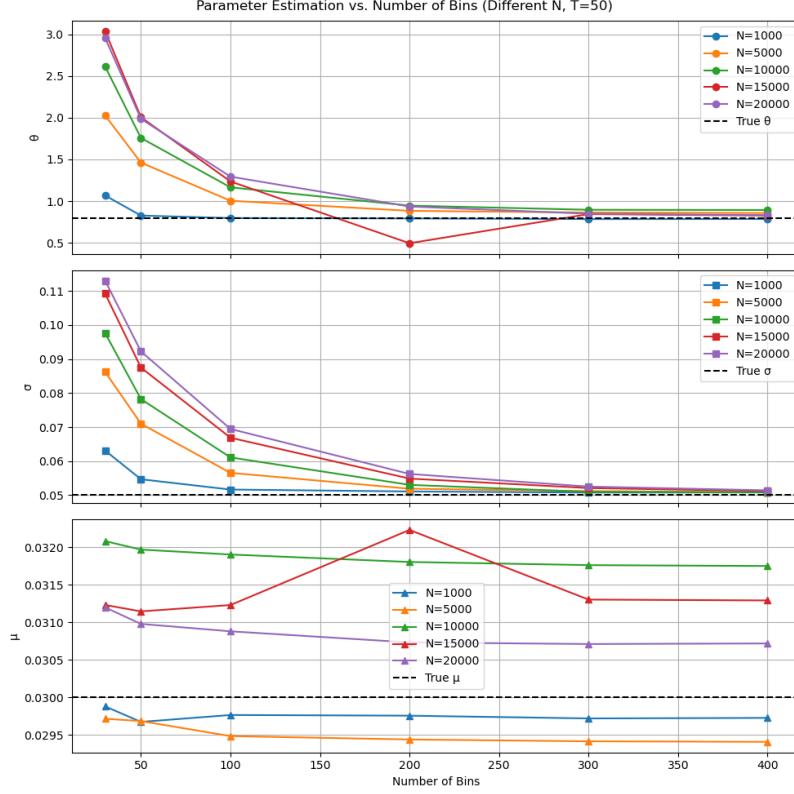


Figure 6.6: Fix $T = 50$, set different $N = 1000, 5000, 10000, 15000, 20000$

For coarser time subdivisions, we observe faster convergence and smaller estimation errors. This phenomenon is expected: with more discretized data, the data structure itself better aligns with the discrete nature of Markov processes. In such cases, each data point tends to convey more significant state transition information. Therefore, as long as the estimation error remains within an acceptable range, we prefer data that is more representative rather than overly detailed.

At the same time, we notice in Figure 6.6 that when the spatial discretization degree is 200, the curve corresponding to $N = 15000$ exhibits a relatively large estimation error for the parameters θ and μ . Similar anomalies have also been observed in other experiments. These occasional spikes in error suggest potential instability in our method under certain conditions. We plan to explore the causes of these deviations in future work.

Up to this point, our algorithm has relied on uniformly spaced lattice points. In the next stage, we investigate the relationship between non-uniform grids and data distribution. We suggest that the Tavella and Randall grid design offers a more effective approach to discretizing the state space. This method ensures that regions with denser actual data points are also assigned higher resolution in the grid, which aligns with empirical intuition. However, a detailed implementation and error analysis of non-uniform grids have not yet been included in this paper and remain part of our future research plan.

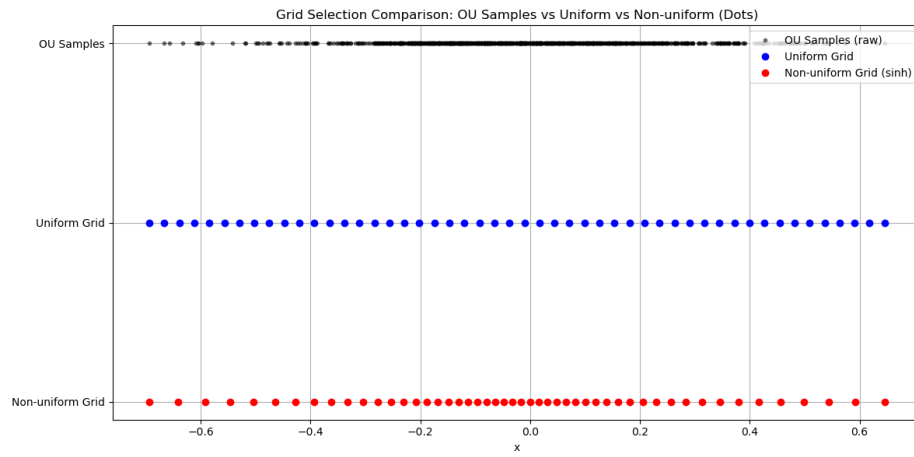


Figure 6.7: Comparison of the actual data with grid design

Chapter 7

Conclusion

In conclusion, this report introduces a framework for using continuous-time Markov chains to approximate diffusion processes. We highlight three main areas of focus in this report: the derivation of convergence conditions, error analysis with grids design, and the parameter estimation algorithm. This article has studied and proved all the three issues mentioned above. Now let's summarize them one by one.

First of all, we conduct a literature review on the background of the problem and recent related papers and state an overview of relevant operator theory with the background of the diffusion operator.

In the methodology section, we describe the process of deriving the convergence conditions for CTMC approximations, emphasizing the needs to satisfy a specific conditional matrix result when setting up these approximations. According to this condition, we can always find a unique continuous-time Markov chain to approximate specific diffusion process. Furthermore, the convergent condition for diffusions with jump terms are also discussed in the following part. In the error analysis section, we perform an error analysis for the CTMC approximation operator, showing that the error primarily depends on the number of grids and the chosen grid design method. Consequently, this leads us to a further discussion in different grid design (spatial discretization) methods and we have proved that the conditions for reducing errors by the non-uniform grid design method.

In the algorithm section, we created an algorithm for parameter estimation for different stochastic process through Markov chain operator approximation. Through this algorithm, we can obtain the estimation of parameters efficiently and quickly under the given random process data, it is equipped with a fast convergent rate. Meanwhile, we presented the image simulation and numerical error analysis of the algorithm, which fully demonstrates the effectiveness and value of the algorithm and is worthy of our further application in actual data in the future.

However, this final report does not yet fully finished. In the future plan, we aim to refine this algorithm into a useful tool for model parameter estimation in related fields. Additionally, there still left many interesting open questions within this topic that warrant further investigation. These include boundary value problems in diffusion processes, high-dimensional diffusion approximation, error analysis in diffusion with jumps.

Appendix

7.1 Appendix A

• Derivation of Diffusion Operator with Jump Component

We begin by reviewing the definition of the Poisson point process, which is expressed as:

$$N(dt, dx) = \sum_{n=1}^{\infty} \delta_{\xi_n}(dt, dx) \quad (7.1)$$

where δ_{ξ} is the counting measure driven by a coin tossing random process ξ . The process $N(dt, dx) \sim \text{Poi}(\mu(dt, dx))$ is a two-dimensional Poisson process, where the intensity measure is given by $\mu(dt, dx) = q(t, dx)dt$, and $\lambda t := \mu([0, t] \times \mathbb{R})$ denotes the expected number of jumps up to time t . Additionally, we can define the compensator of the Poisson measure as:

$$\mathbb{E}[N(dt, dx)] = q(t, dx)dt \quad (7.2)$$

where intensity $q(t, dx)$ means the average jump number with jump size dx until time t . If the intensity is a constant, we can define Lévy measure as $\nu(dx) = q(t, dx)$, representing the distribution of jump amplitudes. In the more general case where $q(t, dx)$ is stochastic and state-dependent, we write the intensity as $\nu(dx) = \lambda(X_t)q(t, dx)$, linking the state-dependent jump rate $\lambda(X_t)$ with the distribution of jump sizes. Thus, the pure jump component in the process is now well characterized.

In Section 3.2, we previously derived the family of operators for a standard diffusion process. We now extend this framework to include jump components. Consider the following stochastic differential equation:

$$dX_t = \mu(X_t)dt + \sigma(X_t)dW_t + \int_{\mathbb{R}} h(X_t, y)N(dt, dy), \quad (7.3)$$

where the function $h(t, x)$ can be seen as the realized jump size coefficient at time t with Poisson jump x . In addition, we can divided the jump term into two parts: one with finite activity (big jumps), and the other with infinite activity (small jumps). It is also called the Lévy–Itô formula as:

$$dX_t = \mu(t, X_t)dt + \sigma(t, X_t)dW_t + \int_{|x|>\epsilon} h_1(t, x)N(dt, dx) + \int_{|x|\leq\epsilon} h_2(t, x)\tilde{N}(dt, dx) \quad (7.4)$$

The only difference from the classical diffusion semigroup operator lies in the inclusion of a jump operator to account for the discontinuous evolution of the process. Therefore, the infinitesimal generator $\mathcal{G}$ introduced in Equation (3.3) is now extended to:

$$\mathcal{G}f(x) = \mu(x)\frac{\partial f}{\partial x} + \frac{1}{2}\sigma^2(x)\frac{\partial^2 f}{\partial x^2} + \int_{\mathbb{R}} [f(x + h(x, y)) - f(x) - h(x, y)f'(x)]\nu(dy) \quad (7.5)$$

Proof. By Itô formula, consider a arbitrarily differentiable function $f(x)$

$$df(X_t) = \frac{\partial f}{\partial x}(X_{t-})dX_t + \frac{1}{2} \frac{\partial^2 f}{\partial x^2}(X_{t-})d[X_t, X_t] + \int_{\mathbb{R}} [f(X_{s-} + h(X_t, y)) - f(X_{s-})] N(ds, dy) \quad (7.6)$$

$$= \mu(X_t) \frac{\partial f}{\partial x}(X_{t-})dt + \sigma(X_t) \frac{\partial f}{\partial x}(X_{t-})dW_t + \frac{1}{2} \sigma^2(X_t) \frac{\partial^2 f}{\partial x^2}(X_{t-})dt \quad (7.7)$$

$$+ \frac{\partial f}{\partial x}(X_{t-}) \int_{\mathbb{R}} h(X_t, y) N(dt, dx) + \int_{\mathbb{R}} [f(X_{s-} + h(X_t, y)) - f(X_{s-})] N(ds, dy) \quad (7.8)$$

$$= \mu(X_t) \frac{\partial f}{\partial x}(X_{t-})dt + \sigma(X_t) \frac{\partial f}{\partial x}(X_{t-})dW_t + \frac{1}{2} \sigma^2(X_t) \frac{\partial^2 f}{\partial x^2}(X_{t-})dt \quad (7.9)$$

$$+ \int_{\mathbb{R}} \left[f(X_{s-} + h(X_t, y)) - f(X_{s-}) + \frac{\partial f}{\partial x}(X_{t-})h(X_t, y) \right] N(ds, dy) \quad (7.10)$$

Take expectation on both sides, we have

$$\mathbb{E}[df(X_t)] = \left(\mathbb{E}[\mu(X_t) \frac{\partial f}{\partial x}] + \frac{1}{2} \mathbb{E}[\sigma^2(X_t) \frac{\partial^2 f}{\partial x^2}] \right) dt \quad (7.11)$$

$$+ \mathbb{E} \left[\int_{\mathbb{R}} \left[f(X_{s-} + h(X_t, y)) - f(X_{s-}) + \frac{\partial f}{\partial x} h(X_t, y) \right] N(ds, dy) \right] \quad (7.12)$$

$$:= \mu(X_t) \frac{\partial f}{\partial x} + \frac{1}{2} \sigma^2(X_t) \frac{\partial^2 f}{\partial x^2} + \int_{\mathbb{R}} \left[f(x + h(x, y)) - f(x) - h(x, y) \frac{\partial f}{\partial x} \right] \nu(dy) \quad (7.13)$$

□

This completes our proof.

Remark 7.1. *Lévy-Itô formula defined as:*

$$df(X_t) = \mu(X_t) \frac{\partial f}{\partial x} dt + \sigma(X_t) \frac{\partial f}{\partial x} dW_t + \frac{1}{2} \sigma^2(X_t) \frac{\partial^2 f}{\partial x^2} dt \quad (7.14)$$

$$+ \int_{|x| > \epsilon} [f(X_{s-} + h_1(X_t, y)) - f(X_{s-})] N(ds, dy) + \int_{|x| \leq \epsilon} [f(X_{s-} + h_2(X_t, y)) - f(X_{s-})] \tilde{N}(ds, dy) \quad (7.15)$$

$$+ \int_{|x| \leq \epsilon} \left[f(X_{s-} + h_2(X_t, y)) - f(X_{s-}) - h_2(X_t, y) \frac{\partial f}{\partial x}(X_{s-}) \right] ds \nu(dy) \quad (7.16)$$

7.2 Appendix B

• **Proof of Remark 4.2** The following lemma is an important result that helps in developing error estimates. It explains the idea that two semigroups are 'close' if their corresponding infinitesimal generators are also 'close'. The error function holds that

$$\epsilon(f) = \sup_{t \in [0, T]} \|Pf - P_n f\|_n \leq \int_0^T C_f(s) ds \cdot h(n, x_i^h). \quad (7.17)$$

where the semigroup convergence only related to its generator convergence.

Proof. By Lemma 6.2 we have the equation

$$P_t f - P_t^{(n)} f = \int_0^t P_{t-s}^{(n)} \left(Q^{(n)} P_s f - \mathcal{L} P_s f \right) ds \quad (7.18)$$

Thus, the triangle inequality yields that

$$\|P_t f - P_t^{(n)} f\|_n \leq \int_0^t \|P_{t-s}^{(n)} (Q^{(n)} P_s f - \mathcal{L} P_s f)\|_n ds \quad (7.19)$$

$$\leq \int_0^t \|Q^{(n)} P_s f - \mathcal{L} P_s f\|_n ds \quad (7.20)$$

$$\leq \int_0^T C_f(s) ds \cdot h(n, x_i^h) \quad (7.21)$$

□

Lemma 7.2. Consider two semigroups $T_A(t)$ and $T(t)$ with generator A and $A + B$, we have

$$T(t)f = T_A(t)f + \int_0^t T_A(t-s)AT(s)f ds \quad (7.22)$$

Lemma 7.3. By fundamental theorem of calculus we can get

$$P_t f - P_t^{(n)} f = \int_0^t \frac{\partial}{\partial s} \left(P_{t-s}^{(n)} P_s f \right) ds = F(t) - F(0) \quad (7.23)$$

Therefore, we only have to consider a general differential on semigroup

$$\frac{\partial}{\partial s} \left(P_{t-s}^{(n)} P_s f \right) = Q^{(n)} P_{t-s}^{(n)} P_s f - \mathcal{L} P_{t-s}^{(n)} P_s f \quad (7.24)$$

$$= P_{t-s}^{(n)} \left(Q^{(n)} P_s f - \mathcal{L} P_s f \right) \quad (7.25)$$

7.3 Appendix C

• Introduction to Kalman Filter and Markov Chain Monte Carlo for Parameter Estimation

Here, we provide a simplified algorithmic overview to help readers review two widely used parameter estimation methods.

We begin with a brief review of the standard Kalman Filter.

The conditional random variable is defined as:

$$X_{n|n-1} := X_n \mid X_1, \dots, X_{n-1} \quad (7.26)$$

Therefore, we can define conditional expectation and variance in a similar notation:

$$\hat{x}_{n|n-1} = \mathbb{E}[X_{n|n-1}] = \mathbb{E}[X_n \mid X_1, \dots, X_{n-1}] \quad (7.27)$$

$$P_{XX(n|n-1)} = \text{Var}[X_{n|n-1}] = \text{Var}[X_n \mid X_1, \dots, X_{n-1}] \quad (7.28)$$

We also define the latent (unobserved) state variable x_t and the observed variable z_t through the following state-space model:

$$\begin{cases} x_t = Ax_{t-1} + B + v_t \\ z_t = Hx_t + C + u_t \end{cases} \quad (7.29)$$

where v_t and u_t denote the process and observation noise, respectively, typically assumed to be independent Gaussian noise with zero mean and known covariance matrices. Based on the above set up, we can summarize the Kalman filter algorithm in the following steps (see next page, Algorithm 2).

Lautier, Javaheri and Galli (2003) in [5] proposed a parameter process estimation method for financial models using the Kalman filter framework. They consider a state vector of the form $\mathbf{x}_t = \begin{pmatrix} x_t \\ \Theta_t \end{pmatrix}$ where Θ_t represents all the time-varying parameters, and the observation variable is given by $\mathbf{z}_t = \text{stock price}$. This framework can be incorporated into specific financial models, such as the Heston stochastic volatility model:

$$\begin{cases} dS_t = \mu S_t dt + \sqrt{v_t} S_t dW_t^{(1)} \\ dv_t = \kappa(\theta - v_t)dt + \sigma \sqrt{v_t} dW_t^{(2)} \end{cases} \quad (7.30)$$

with the price under this model can be expressed through a pricing function of the form:

$$F(S_t, v_t, t, T) = S_t e^{-C(t) \frac{1 - e^{\kappa(T-t)}}{\kappa} + B(T-t)} \quad (7.31)$$

To apply the Kalman filter, all parameters in the model are treated as latent components embedded within the state variable. The filter dynamics can be expressed as:

$$\mathbf{x}_t = \begin{pmatrix} \log(S_t) \\ v_t \end{pmatrix} = \begin{pmatrix} (\mu - \frac{1}{2}\sigma_s^2)\Delta t \\ \kappa\theta\Delta t \end{pmatrix} + \begin{pmatrix} 1 & -\Delta t \\ 0 & -\kappa\Delta t \end{pmatrix} \begin{pmatrix} \log(S_{t-1}) \\ v_{t-1} \end{pmatrix} + \begin{pmatrix} \sigma_s^2\Delta t & \rho\sigma_s\sigma_v \\ \rho\sigma_v\sigma_s & \sigma_v^2\Delta t \end{pmatrix} \quad (7.32)$$

Algorithm 2: Standard Kalman Filter

Input: Initialize assumption parameters**Output:** Estimated parameters**1 Initialize the algorithm at $t = 0$ using unconditional moments**

$$\hat{x}_{0|0} = \mathbb{E}[x_{0|0}]$$

$$P_{xx(0|0)} = \text{Var}[x_{0|0}]$$

2 Predict the next state vector For $t = 1, \dots, T$:

$$\hat{x}_{t|t-1} = A\hat{x}_{t-1|t-1} + B$$

$$P_{xx(t|t-1)} = AP_{xx(t-1|t-1)}A^T + Q_{t|t-1}$$

3 Forecast the measurement system conditionally

$$\hat{z}_{t|t-1} = H\hat{x}_{t|t-1} + C$$

$$P_{zz(t|t-1)} = HP_{xx(t|t-1)}H^T + R_{t|t-1}$$

4 Calibrate the estimate of state variablesGiven the real value z_t , the prediction error for calibration is $z_t - \hat{z}_{t|t-1}$.

$$K_t = P_{xz(t|t-1)}P_{zz(t|t-1)}^{-1} = P_{xx(t|t-1)}H^TP_{zz(t|t-1)}^{-1}$$

The updating state variables is done using conditional moments:

$$\hat{x}_{t|t} = \hat{x}_{t|t-1} + K_t(z_t - \hat{z}_{t|t-1})$$

$$P_{xx(t|t)} = P_{xx(t|t-1)} - K_tP_{zz(t|t-1)}K_t^T$$

and

$$\mathbf{z}_t = x_t \frac{1 - e^{\kappa(T-t)}}{\kappa} + B(T-t) + \epsilon_t \quad (7.33)$$

where ϵ_t represents measurement noise. In this framework, all parameter information is embedded in the latent state vector x_t . At each iteration step, parameter estimation is achieved by maximizing the log-likelihood function:

$$l(\theta) = \sum_k \log p(z_k | z_1, \dots, z_{k-1}), \quad (7.34)$$

which is naturally obtained as a by-product of the Kalman filter recursion.

Hence, the key idea behind applying the Kalman filter in parameter estimation is to treat both the latent state and the model parameters as part of a unified state-space structure, enabling efficient joint estimation through likelihood maximization.

Next, let us recap the Markov Chain Monte Carlo (MCMC) parameter estimation method,

which is one of the most important algorithms in applied statistical and computational fields. This approach originates from the Bayesian estimation framework, where we assume the model parameters follow some unknown probability distribution. Given a set of observation data, our goal is to generate the posterior distribution $\pi(\Theta | Y_{\text{data}})$ based on a specified initial assumed distribution $\pi(\Theta)$. According to Bayes' theorem, the posterior distribution can be expressed as:

$$\mathbb{P}(\Theta | Y_{\text{data}}) = \frac{\mathbb{P}(Y_{\text{data}} | \Theta)\pi(\Theta)}{\mathbb{P}(Y_{\text{data}})} = \frac{\mathbb{P}(Y_{\text{data}} | \Theta)\pi(\Theta)}{\int \mathbb{P}(Y_{\text{data}} | \Theta)\pi(\Theta)d\Theta} \quad (7.35)$$

However, how can we result the best estimated distribution of parameter Θ ? The main challenge is typically intractable due to the unknown form of the normalizing constant in the denominator. The integral often lacks an explicit analytical expression. Therefore, we employ the MCMC method to approximate this sampling process, which is particularly useful for estimating dynamic parameter distributions.

MCMC algorithm aims to approximate expectations with respect to the posterior distribution by solving the marginal distribution $\mathbb{P}(f) = \int f(x)p(x)dx$ inference problem, where is indeed solving the problem sampling from a complex probability distribution. The core idea of MCMC methods is to construct an ergodic Markov chain whose stationary distribution is the desired posterior. By simulating the chain for a sufficiently long time, we obtain a set of samples that can be used to approximate expectations and perform inference.

The ergodicity property of the Markov chain ensures convergence to the stationary distribution, and it satisfies:

$$\sum_{i=1}^d f(i)\pi(i) = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T f(x_t) \quad (7.36)$$

where $\pi(i)$ is the target stationary distribution.

Based on the above set up, we can summarize the MCMC algorithm in the following steps:

Algorithm 3: Markov Chain Monte Carlo (MCMC) via Metropolis-Hastings

Input: initial distribution of parameters $\pi(\theta \mid y)$ and proposal kernel $q(\cdot \mid \cdot)$ **Output:** posterior target distribution of parameter θ **1 Initialize:** Select initial assumption parameter state $\Theta_{(0)}$ **2 while** $i \leq N$ **do****3 Propose:** Generate candidate θ^* from proposal distribution:

$$\theta^* \sim q(\cdot \mid \theta^{(i-1)})$$

4 Acceptance Ratio: Compute the Metropolis-Hastings acceptance probability:

$$\alpha = \min \left(1, \frac{\pi(\theta^* \mid Y) q(\theta^{(i-1)} \mid \theta^*)}{\pi(\theta^{(i-1)} \mid Y) q(\theta^* \mid \theta^{(i-1)})} \right)$$

5 Accept/Reject: Draw $u \sim \mathcal{U}(0, 1)$.**6 if** $u < \alpha$ **then****7**Set $\theta^{(i)} = \theta^*$ Accept the candidate**8 else****9**Set $\theta^{(i)} = \theta^{(i-1)}$ Reject and stay at current state**10**Update iteration $i \rightarrow i + 1$

That is the end of Appendix.

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